Technische Universität Braunschweig

Development and evaluation of predictive machine learning methods for forecasting demand for battery electric vehicles

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Table of Contents

Motivation

1. **EU mandate:** Zero-emission new car sales by 2035

2. BEV's central to transport policy, energy systems planning, infrastructure investment

3. COVID 19's structural break with BEV sales ↑ 43%, total car sales ↓ 16% in 2020

4. Post-2020 market has seen faster adoption and different dynamics

Problem: Structural break poses misleading forecasts

Solution: Need for robust, updated forecasting approaches

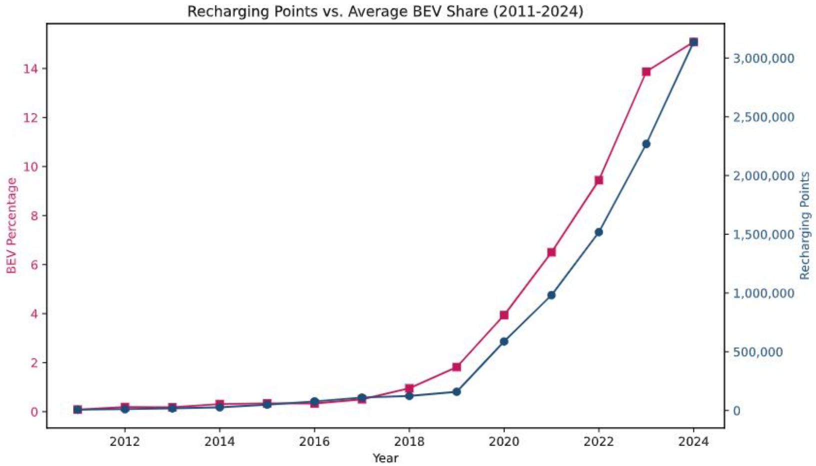


Table 2.2: Comparison of studies across key aspects

Study	G1	G2	G3	G4	G5	G6
Sierzchula et al. (2014) [Sie+14]	●	○	○	○	○	○
Zhang et al. (2016) [Zha+16]	○	○	○	◐	○	○
Zhang et al. (2017) [Zha+17]	○	○	○	◐	○	◐
Wang et al. (2019) [WGW19]	○	○	○	○	◐	◐
Lieven & Hügler (2021) [LH21]	●	◐	◐	○	○	○
Domarchi & Cherchi (2023) [DC23]	●	◐	◐	●	○	◐
Möring-Martínez et al. (2024) [MSJ24]	●	○	●	○	○	◐
Kemala et al. (2024) [KSP24]	●	◐	◐	●	○	◐
This study (2026)	●	●	●	●	●	●

○ not fulfilled ◐ partially fulfilled ● fully fulfilled

G1 = Geographic scope of study (single-country to multiple countries), **G2** = Incorporation of structural breaks, **G3** = Temporal coverage (short to long-term history), **G4** = Method comparison (single model to multiple model comparison), **G5** = Use of hybrid or ensemble approaches, **G6** = Validation rigor (in-sample to robust out-of-sample evaluation)

Research Questions

Question 1: Can pre-2020 models predict post-2020 BEV adoption? Do EU countries behave differently after COVID?

Question 2: Does adding 2020–2022 data improve forecasts?

Question 3: Are post-2020-only models more accurate than long-history models?

Question 4: Do EU countries form distinct adoption clusters? Can cluster-specific models outperform a single model?

Research Hypotheses

Hypothesis 1: Structural Instability

- Models trained on 2011–2019 data will give higher prediction error for 2020-2024.

=> Evidence of structural change in market dynamics

Hypothesis 2: Transitional Data Effect

- Extending training to **2022 improves accuracy but bias remains the same**

=> Underestimation due to pre-COVID data dominance

Hypothesis 3: Post-COVID Regime Superiority

- Models trained on **2020–2023 data only will have** highest predictive performance

=> Indicates new market regime.

Hypothesis 4: Stable margaret segmentation

- EU-27 shows a table two-cluster configurations - **Leaders and Followers**, across all training configurations.

=> Clear structure is robust, though individual country membership evolve over time.

Dataset Overview

Coverage
27 EU member states
2011 – 2024 (14 years)
378 observations
Annual frequency
Target: BEV market share

Table 2.1

10 variables across 3 categories		
Category	**Variables**	**Level**
1. Socioeconomic	GDP · CPI · Employment growth	Country-level (unique per country)
2. Infrastructure	Recharging points	
3. Technological	Real range · Battery capacity · AC/DC speed · Available models · Purchase price	EU-level average (same for all)

Table 2.2

Workflow of the Analysis

Data Cleaning & Imputation	Dimensionality Reduction	Clustering	Model Development	Evaluation Metrics
Merge different datasets using **country + year as keys** Imputation of missing values Transformation of skewed variables	**Pearson Correlation:** to identify informative predictors. **VIF analysis:** to check the level of multicollinearity. **PCA:** to overcome collinearity while transforming them into orthogonal component with a VIF=1. Applied only on standardized training data to avoid data leakage	To determine **#of clusters (k):** Elbow method + Silhouette score (for validation) All trials identified **k=2**, consistently. Applied **K-Means** to PCA scores separately for training and test periods in each trial	**9 models** trained and evaluated across **3 trials × 2 clusters** All models trained on **cluster-specific training data only** All predictions **back-transformed** via exp() before evaluation	**MAE:** captures overall prediction & sensitive to extreme outliers. **MedAE:** robust to outliers Both metrics computed in **original BEV% scale** & reported **separately for Leaders and Followers** in each trial

Pre-Processing: Handling missing data

	Log-Linear Interpolation	CAGR Extrapolation	Hybrid Approach
Applied where	For internal gaps where growth is multiplicative	For terminal gaps (missing values at the end of the time series)	For variables with both internal and terminal gaps
Applied to	BEV market share & Purchase Price	Available BEV models & Real Range (km)	AC/DC recharging speed & Battery Capacity
Why/How?	Preserves non-negativity Avoids unrealistic jumps	Uses last 3 years to calculate the missing year	Log-Linear first, then CAGR

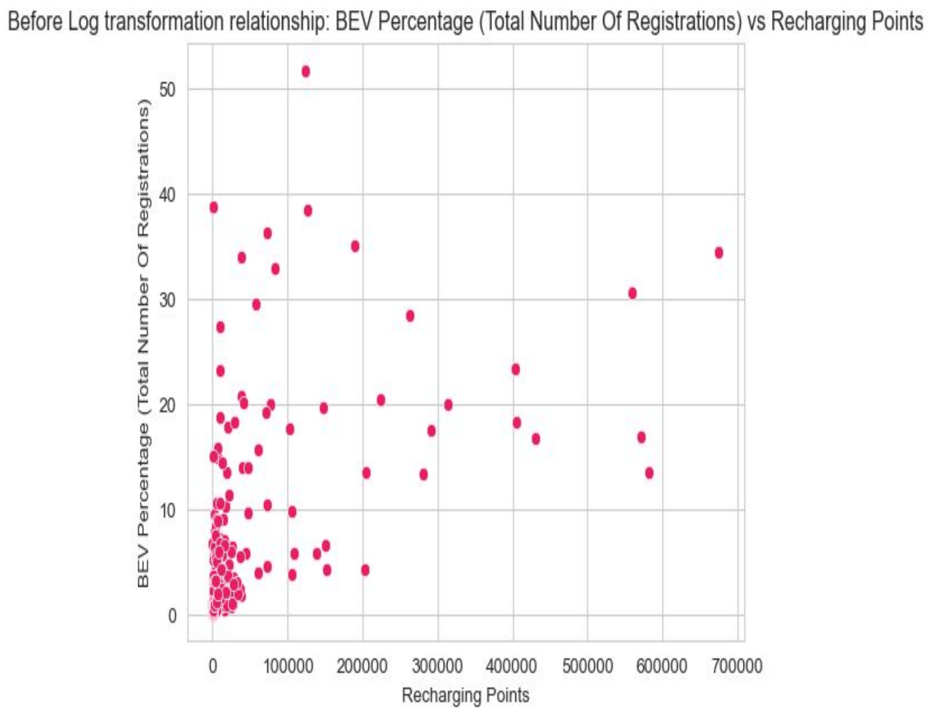
Table 2.3

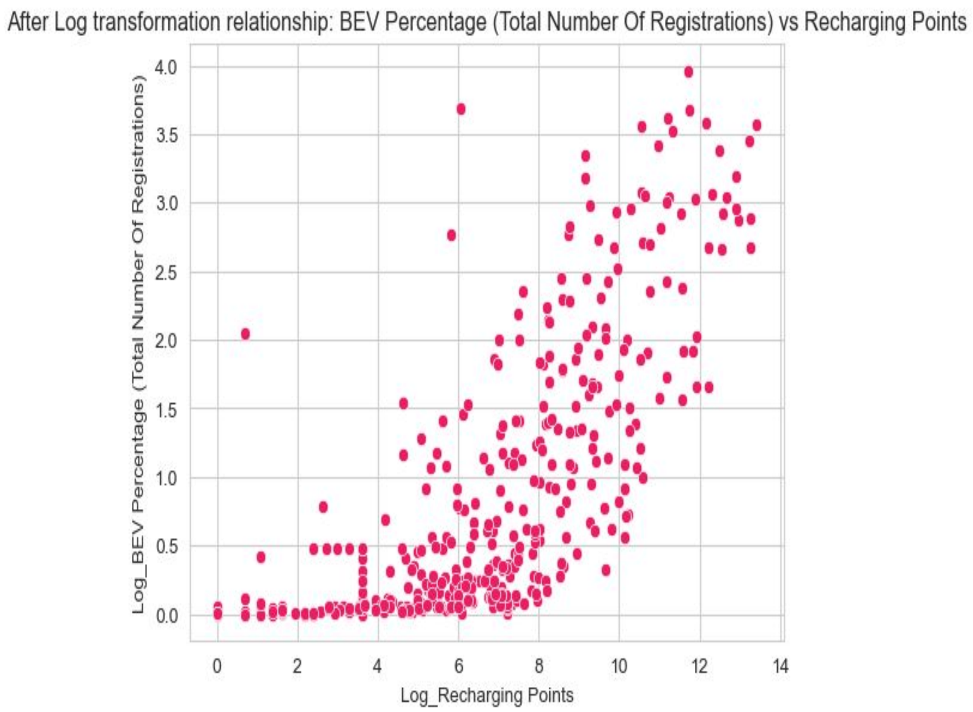
Exploratory Data Analysis: Transformations

Transform variables if |1| < skewness < |1|

Descriptive Statistics			
Skewed Variables	**Skewness Before —> After Transformation**	**Category**	**Transformation**
GDP	5.76 —-> 0.29	High skewness	Log Transform
Recharging Points	5.29 —-> -0.21	High skewness	Log Transform
Available BEV models	1.90 —->0 0.33	High skewness	Log Transform

Exploratory Data Analysis: Effect of Transformations





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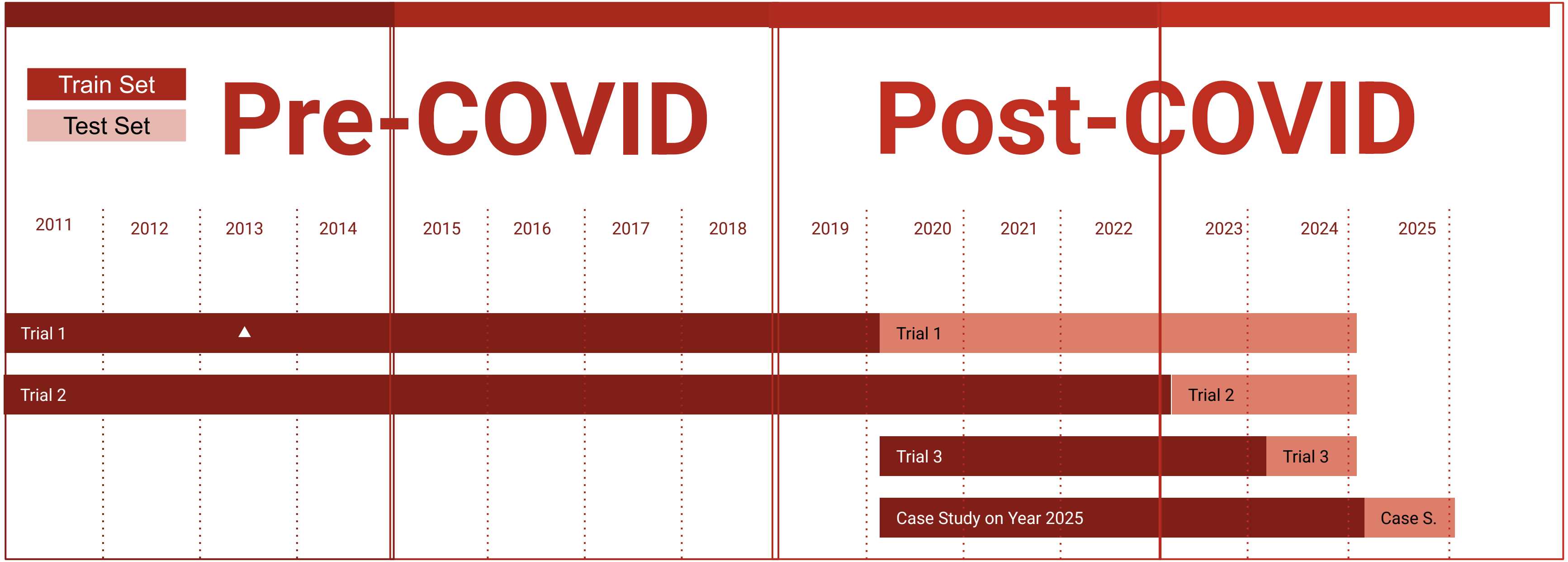
Design of Trials

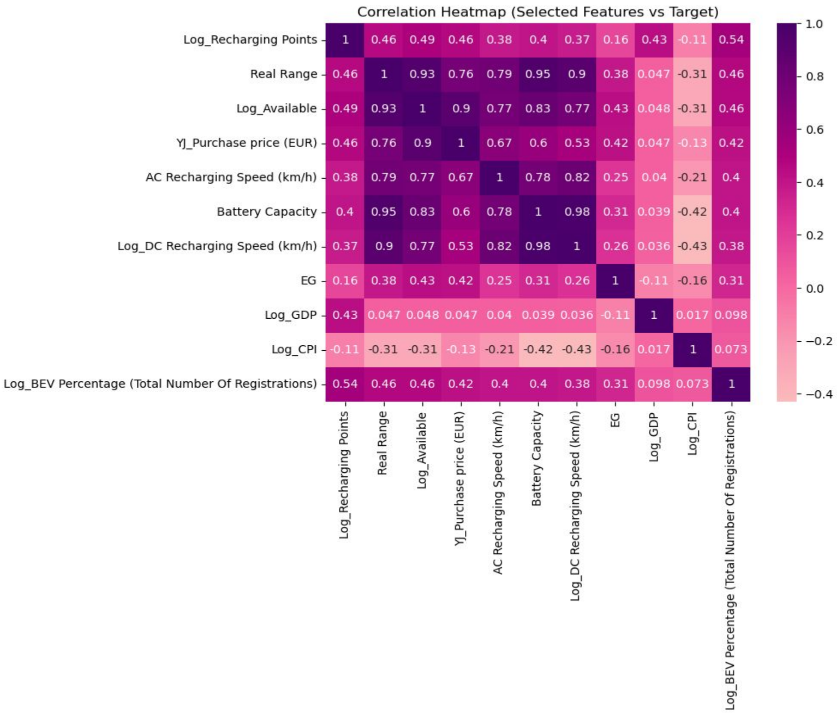


Implementation of Experiments

Feature Selection & Dimensionality Reduction

01	**Pearson Correlation**	• Compute correlation of each feature with the target. • Findings: Infrastructure & technological variables show highest correlation.
02	**VIF Analysis**	• Quantify multicollinearity. • Findings: Real Range, Battery Capacity & DC Charging Speed are all severely collinear.
03	**PCA**	• Transform 10 features into principal components. • PCA fitted on train data only to avoid data leakage.

Correlation & VIF analysis for Trial 1



	Feature	VIF
0	Log_Recharging Points	17.045261
1	Real Range	537.361825
2	Log_Available	115.978308
3	YJ_Purchase price (EUR)	4.666438
4	AC Recharging Speed (km/h)	39.427131
5	Battery Capacity	271.273113
6	Log_DC Recharging Speed (km/h)	131.906094
7	EG	1.782205
8	Log_GDP	56.511499
9	Log_CPI	3.290362

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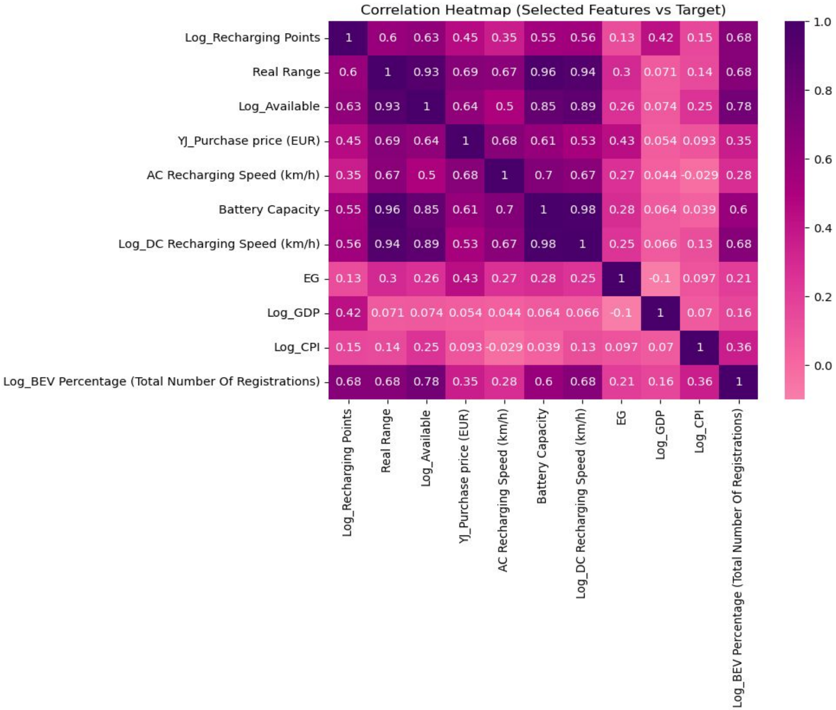
Correlation & VIF analysis for Trial 2



	Feature	VIF
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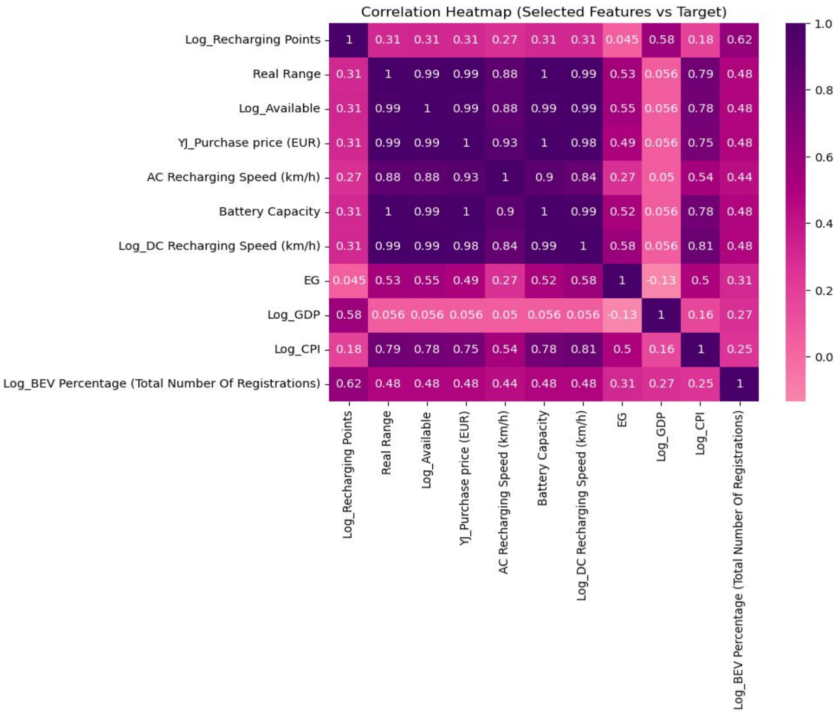
Correlation & VIF analysis for Trial 3



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7	EG	1.782205
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9	Log_CPI	3.290362

PCA

Why was PCA introduced?

1. Multiple predictor variables exhibited **high inter-feature correlations** => confirmed by Pearson correlation matrix

2. **Severe multicollinearity** => confirmed by VIF analysis. Real Range VIF = 537.36, Battery Capacity = 271.27, DC Recharging Speed = 131.91

3. Simply **dropping collinear variables was not sufficient**.

4. A variable can be **both highly correlated with the target AND highly collinear** with other predictors simultaneously

Number of Principal Components in Each Trial

The number of principal components was determined by the elbow method

Trial Number	Components Retained	Variance Explained
Trial 1 (2011–2019)	3	79.36%
Trial 2 (2011–2022)	3	77.14%
Trial 3 (2020–2023)	2	83.53%

Clustering

- Number of clusters determined using **Elbow Method** and validated using **Silhouette Method.**
- **K means** clustering method was used to cluster the EU 27 countries into 2 distinct groups - **Leaders & Followers.**
- These clusters were based on on 2 decisions -
  - Which countries were stable in both the datasets.
  - Which countries changed cluster in the test dataset

Elbow method to determine the number of clusters for Trial 1

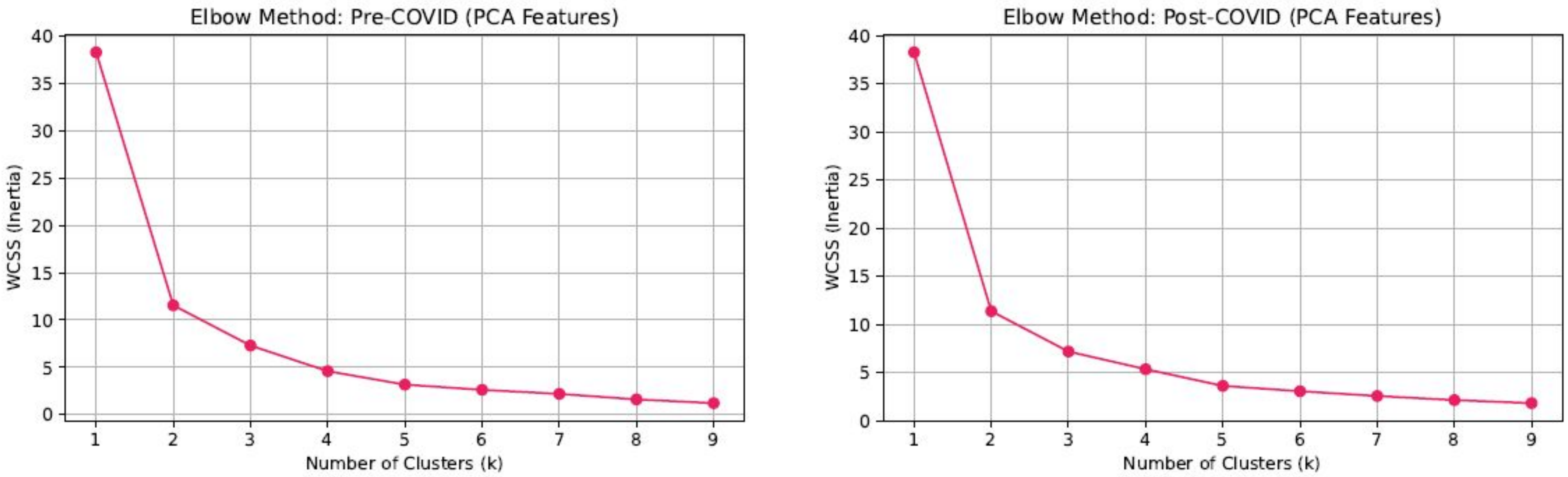


Figure 4.2: Elbow Method - Pre-COVID (left) and Post-COVID (right) PCA Features (Trial 1).

Silhouette method to validate the number of clusters for Trial 1

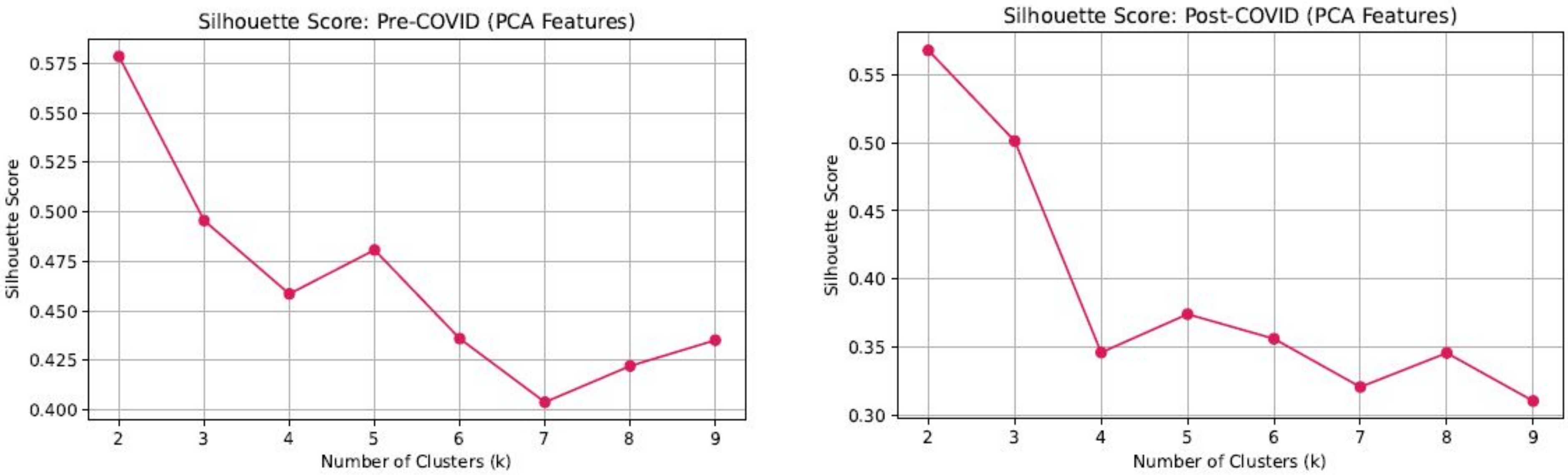


Figure 4.3: Silhoutte Method - Pre-COVID (left) and Post-COVID (right) PCA Features (Trial 1).

Elbow method to determine the number of clusters for Trial 2

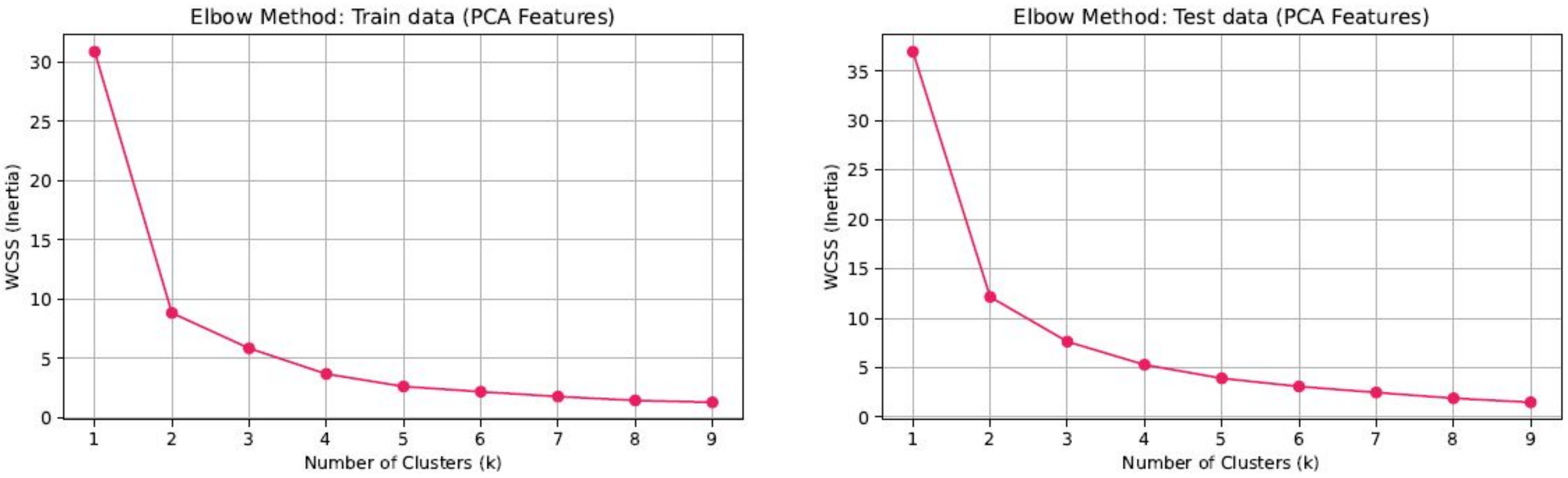


Figure 4.6: Elbow Method - Train Data (left) and Test Data (right) PCA Features (Trial 2).

Silhouette method to validate the number of clusters for Trial 2

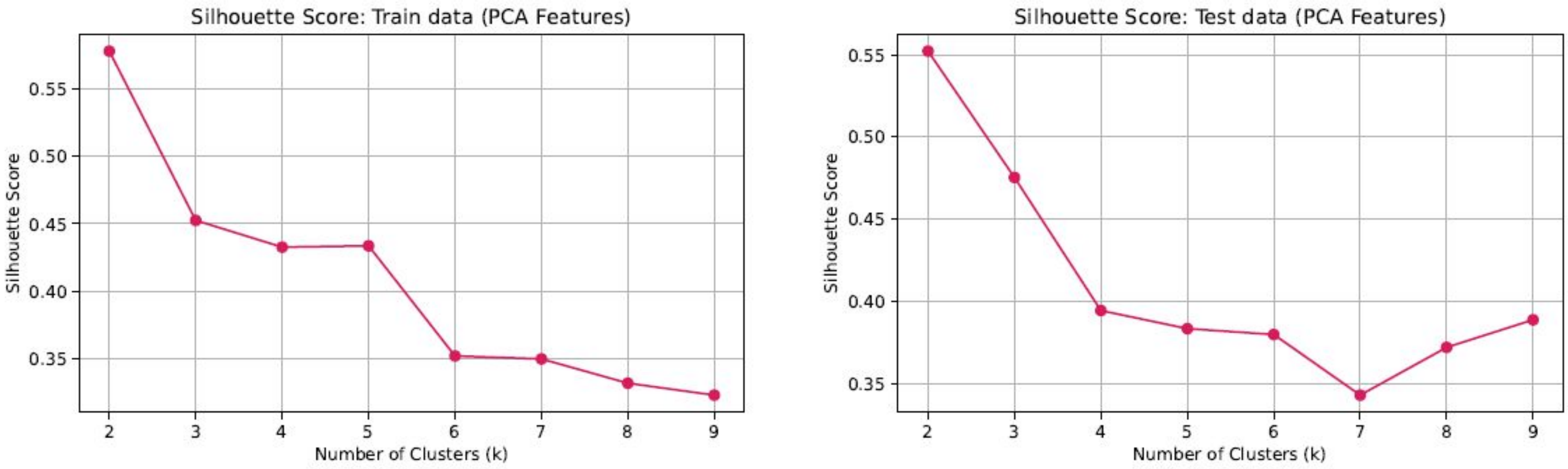


Figure 4.7: Silhoutte Method - Train Data (left) and Test Data (right) PCA Features (Trial 2).

Elbow method to determine the number of clusters for Trial 3

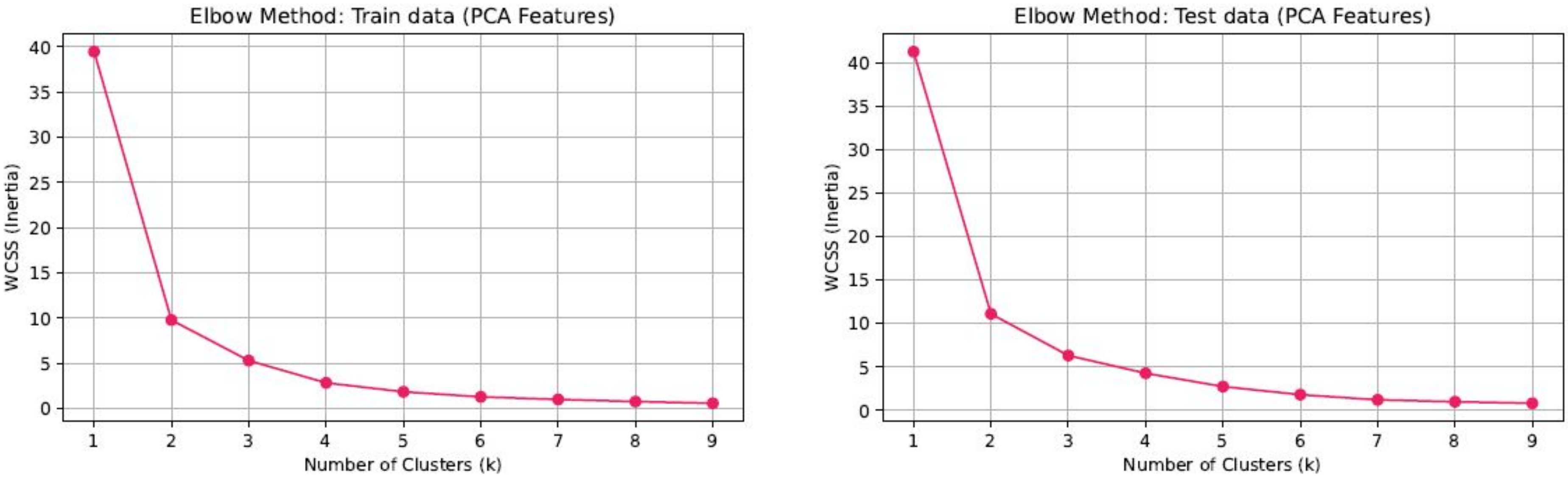
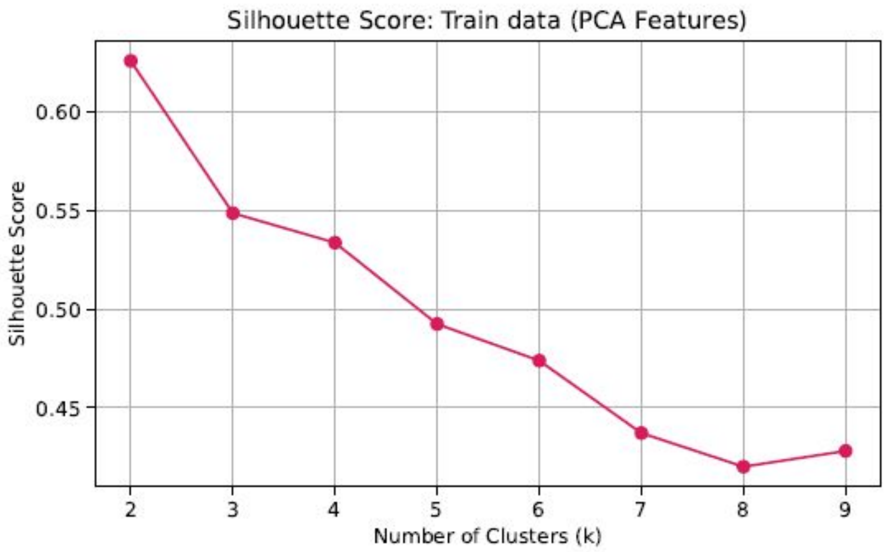


Figure 4.10: Elbow Method - Train Data (left) and Test Data (right) PCA Features (Trial 3).

Silhouette method to validate the number of clusters for Trial 3



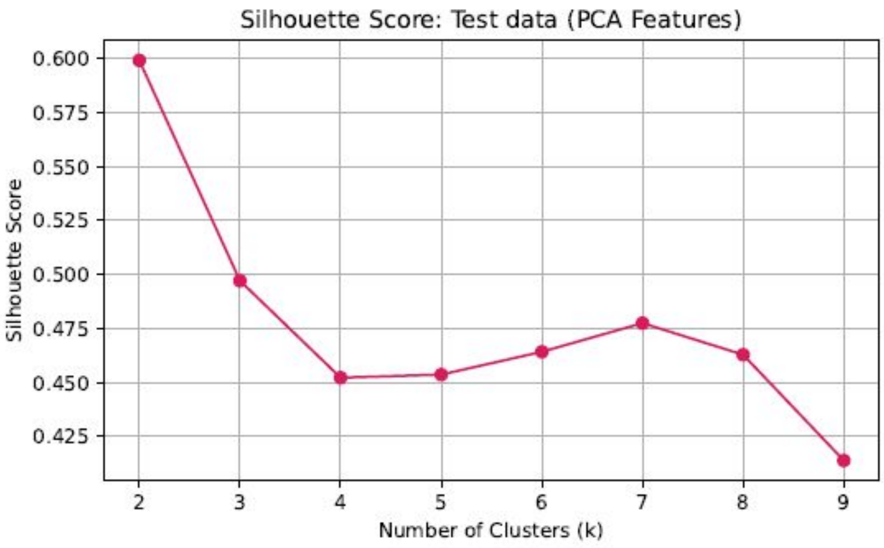
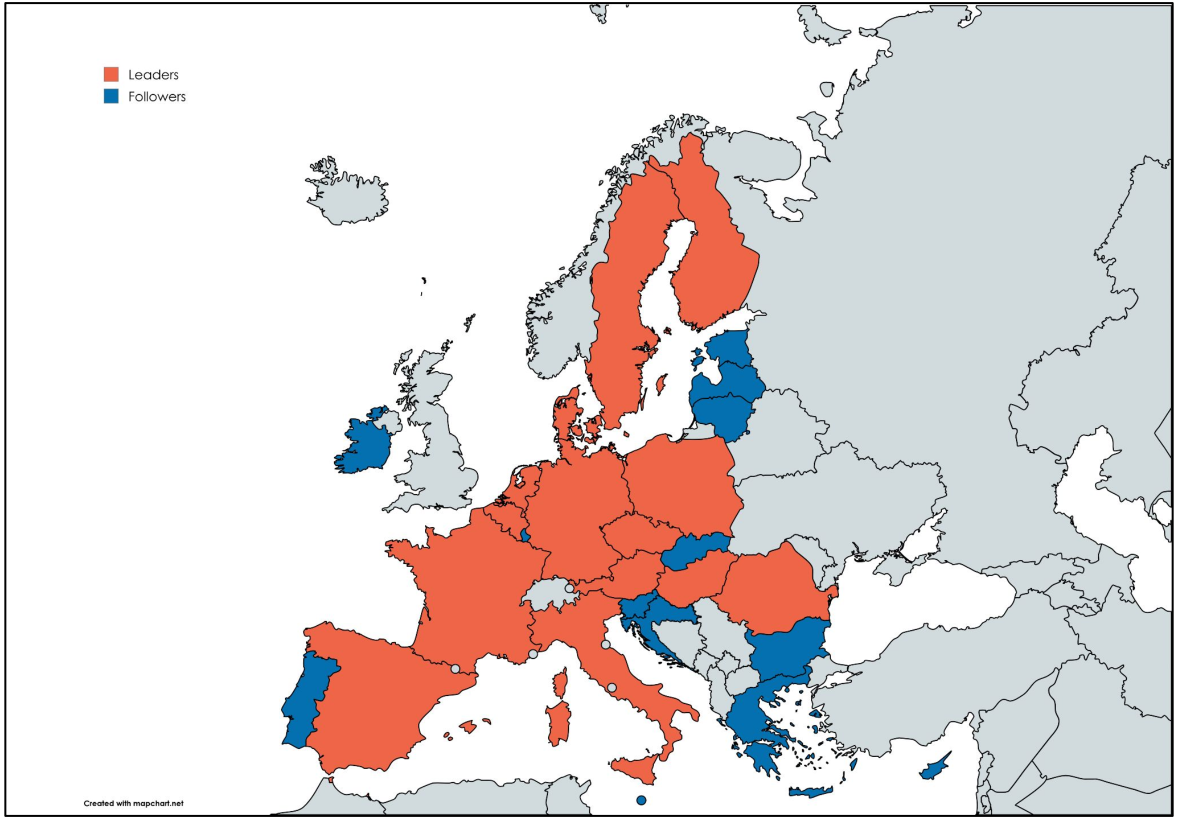
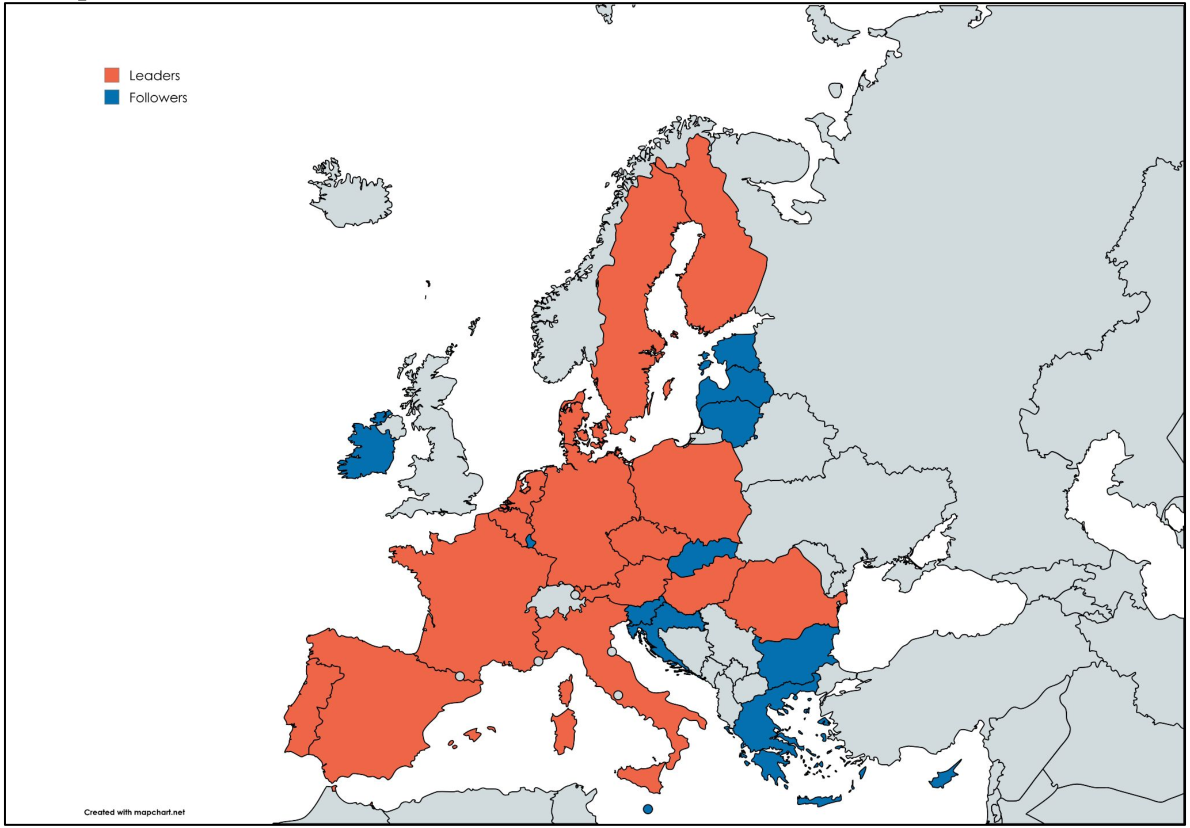


Figure 4.11: Silhoutte Method - Train Data (left) and Test Data (right) PCA Features (Trial 3).

Cluster Composition for Trial 1 and Trial 2



Cluster Composition for Trial 3



Models Used

Logic behind these models were -

- Cover the full spectrum from simplest possible (Linear Regression) to most complex (Ensemble).
- Ensure models with fundamentally different assumptions are included which allows systematic failure mode analysis.

Baseline Models	Time Series Model	Deep Learning	Proposed Stacking Ensemble
Linear Regression (LR)	ARIMAX	LSTM	**Base Model 1:** LSTM **Base Model 2:** ARIMAX **Meta Learner 1:** Elastic Net **Meta Learner 1:** Gradient Boost **Meta Learner 1:** Weighted Average
Support Vector Regression (SVR)			
Random Forest (RF)			
XGBoost			

Pre-Assessment of the Models for Baseline Models

Models	Expected Behavior	Bias/Variance	Assumptions
LR	1. Serves as a simple benchmark model & captures only linear relationships 2. Weak Performance overall	High bias, low variance	1. Linearity 2. Independence of errors 3. No structural breaks
SVR	1. Captures non-linear trends because using RBF kernel. 2. Strong candidate for moderately complex/stable regimes.	1. Medium bias, medium variance. 2. High sensitivity to hyperparameters.	1. Assumes there is no structural break 2. Training & Test set has same distribution.
RF	1. Captures non-linear patterns and a strong ensemble baseline. 2. Strong for heterogeneous data but weak under structural shifts.	1. Low bias, reduced variance via averaging 2. Robust to noise & irrelevant features	1. Training & test data distribution is the same. 2. Cannot predict outside the range of values seen during training
XGBoost	1. Captures non-linear relationships and strong predictive accuracy using boosting method, esp. in tabular data. 2. Sensitive to structural breaks and may overfit the patterns in pre-COVID era	1. Lower bias than RF 2. Higher variance risk without regularization.	1. Training & test data distribution is the same. 2. Sequential residual learning representative across training & test periods.

Pre-Assessment of the Models for Baseline Models

Models	Expected Behavior	Bias/Variance	Assumptions
ARIMAX	1. Captures temporal correlations + exogenous drivers 2. Strong for smooth trends but degrades in structural breaks & expected in post-COVID regime.	1. High bias, low variance 2. Strong interpretability but less flexibility	1. Linear Relationships with exogenous variables 2. Weak exogenous outputs
LSTM	1. Captures non-linear trends and dynamic temporal dependencies 2. Weak under mixed regimes (Trial 2) and risk of overfitting due to limited data	1. Low bias, High variance 2. Sensitive to less data	1. Training set temporal patterns tries to generalise in the test set 2. Assumes there is sufficient data for learning the dynamic patterns
Stacking Ensemble	1. Combines linear trends + non linear trends and acceleration effects as seen in structural breaks. 2.Expected best overall performance and robust in heterogeneous data	1. Balanced bias & variance 2. Reduced error through combination	1. Base Learner errors are weakly correlated 2. Meta Learners can generalise the output

Pre-Assessment of the Models for Stacking Models

Models	Expected Behavior	Bias/Variance	Assumptions
Stacking Ensemble + Elastic Net Meta Learner	1. Linear combination of ARIMAX + LSTM and provides an interpretable stacking 2. Most stable stacking variant and is expected to have a better generalisation on small datasets	1. Moderate bias and low variance 2. Regularisation prevents overfitting	1. Linear relationship between base learner o/p and target 2. Both model contribute useful signals
Stacking Ensemble + Gradient Boost Meta Learner	1. Expects non-linear combination of base learner 2. High flexibility and risk of overfitting in small datasets 3. May outperform Elastic Net if patterns are complex	1. Low bias, High variance	1. Sufficient data for residual learning
Stacking Ensemble + Weighted Averaging Meta Learner	1. Simple interpretation of the base learners output 2. Expected to improve stability by averaging the complementary model errors. 3. Strong stacking baseline approach for small datasets but may underperform for highly non linear patterns	1. Moderate bias and low variance 2. Variance reduced through weighted averaging	1. Both base learners give meaning outputs. 2. Base learners errors are weakly correlated 3.Optimal linear weighting remains stable across the training and test periods.

Model Performance & Hypothesis Validation

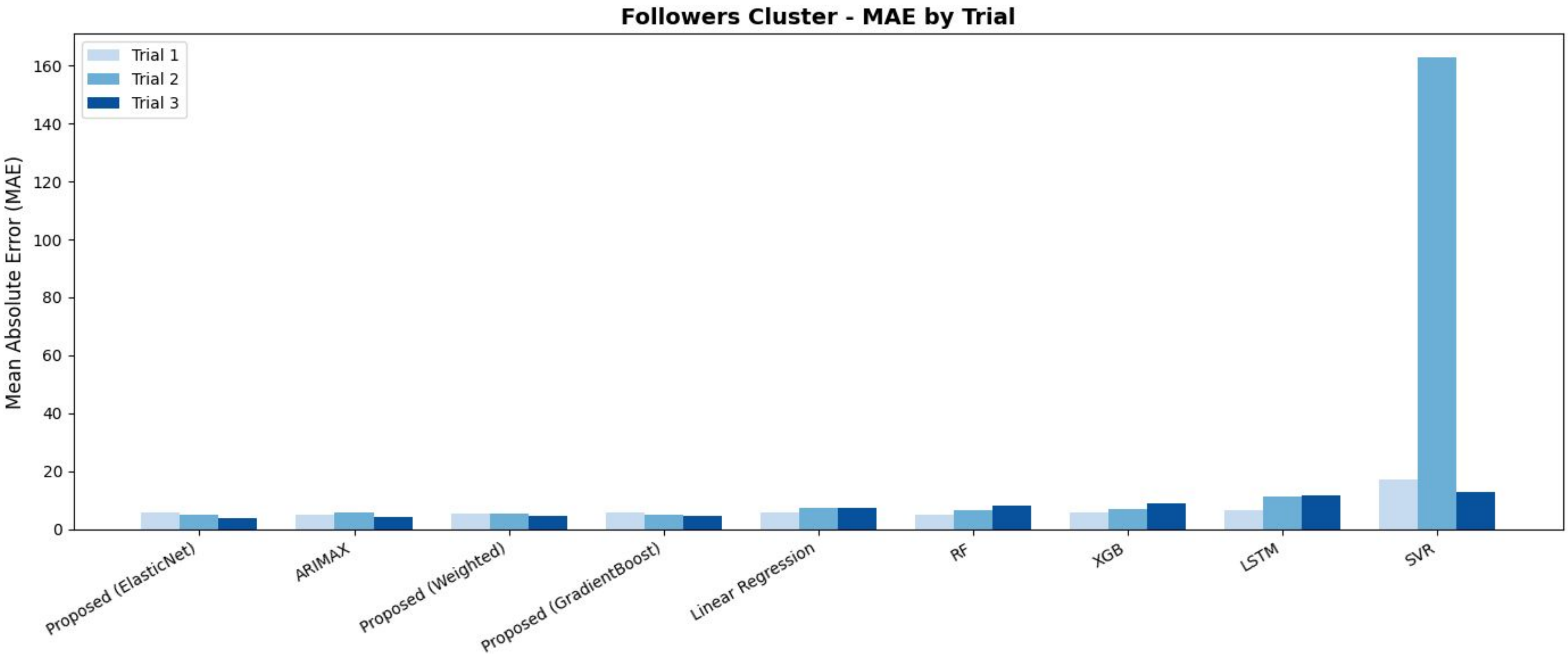
Mean Absolute Error (%) by Model, Cluster, and Trial

Table 4.13: MAE % by Model, Cluster and Trial.

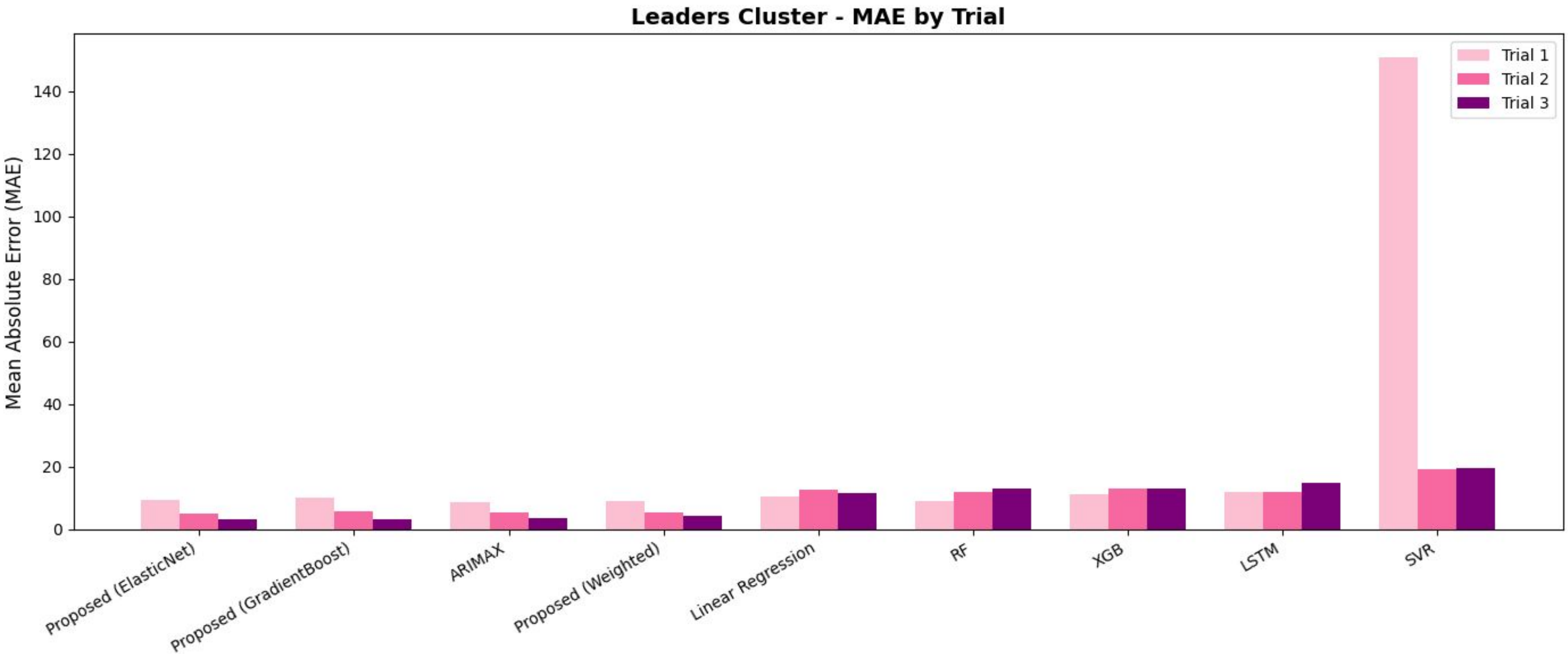
Cluster	Model	Trial 1	Trial 2	Trial 3
Followers	Proposed (ElasticNet)	5.539	5.048	**3.683**
	ARIMAX	4.896	5.723	4.122
	Proposed (Weighted)	5.247	5.445	4.442
	Proposed (GradientBoost)	5.708	5.106	4.508
	Linear Regression	5.918	7.269	7.407
	RF	4.877	6.708	8.172
	XGB	5.868	6.873	8.750
	LSTM	6.328	11.330	11.760
	SVR	17.139	162.848	12.994
Leaders	Proposed (ElasticNet)	9.281	5.071	**2.971**
	Proposed (GradientBoost)	9.938	5.633	3.146
	ARIMAX	8.644	5.298	3.542
	Proposed (Weighted)	9.101	5.467	4.130
	Linear Regression	10.559	12.750	11.649
	RF	8.803	11.830	12.789
	XGB	11.260	12.866	12.821
	LSTM	11.846	11.793	14.620
	SVR	150.710	19.156	19.487

- **H1 CONFIRMED:** Pre-COVID models fail on post-COVID data
- **H2 PARTIALLY CONFIRMED**: Transitional data helps inconsistently
- **H3 STRONGLY CONFIRMED:** Post-COVID training wins
- **H4 CONFIRMED:** Stable two-cluster structure

MAE Visualisation for Follower Cluster



MAE Visualisation for Leader Cluster



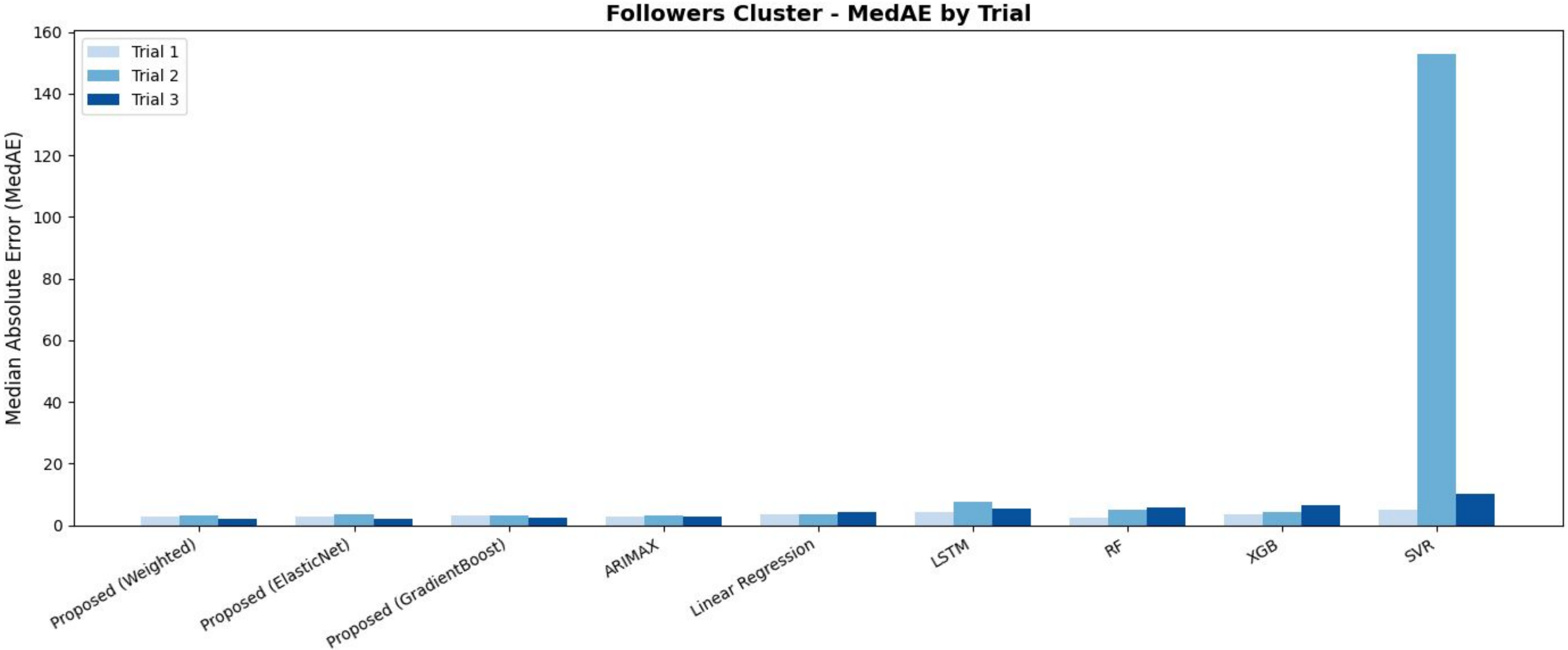
Model Performance

Median Absolute Error (%) by Model, Cluster, and Trial

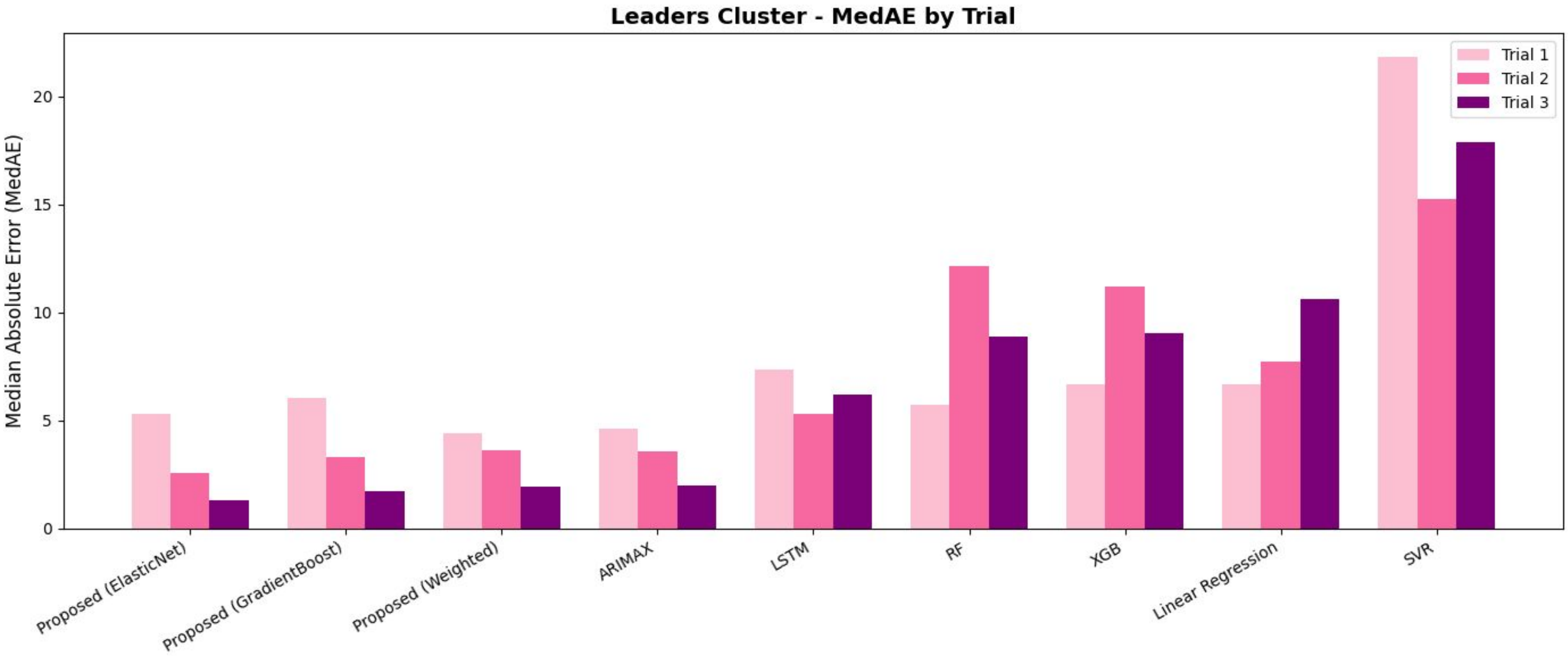
Table 4.14: MedAE % by Model, Cluster and Trial.

Cluster	Model	Trial 1	Trial 2	Trial 3
Followers	Proposed (ElasticNet)	2.9437	3.3987	**2.2079**
	Proposed (Weighted)	2.9485	3.1991	2.1896
	Proposed (GradientBoost)	3.2251	3.1560	2.2462
	ARIMAX	2.7177	3.2305	2.9195
	Linear Regression	3.5502	3.3647	4.2537
	LSTM	4.170	7.453	4.253
	RF	2.4503	4.9279	5.6356
	XGB	3.4803	4.3348	6.6140
	SVR	5.0478	152.8129	10.1787
Leaders	Proposed (ElasticNet)	5.3177	2.5746	**1.2821**
	Proposed (GradientBoost)	6.0523	3.2924	1.7208
	Proposed (Weighted)	4.4244	3.6201	1.9152
	ARIMAX	4.6018	3.5760	1.9570
	LSTM	7.328	5.314	6.165
	RF	5.7276	12.1185	8.8497
	XGB	6.6631	11.1940	9.0384
	Linear Regression	6.6732	7.6936	10.6022
	SVR	21.8152	15.2423	17.8809

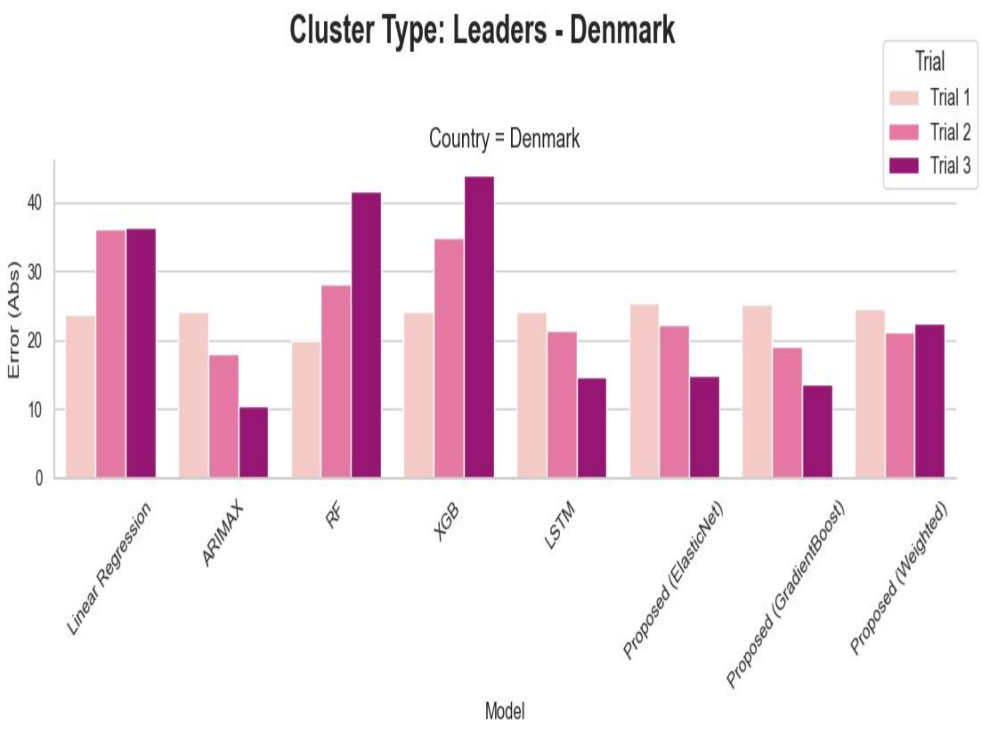
MedAE Visualisation for Followers Cluster

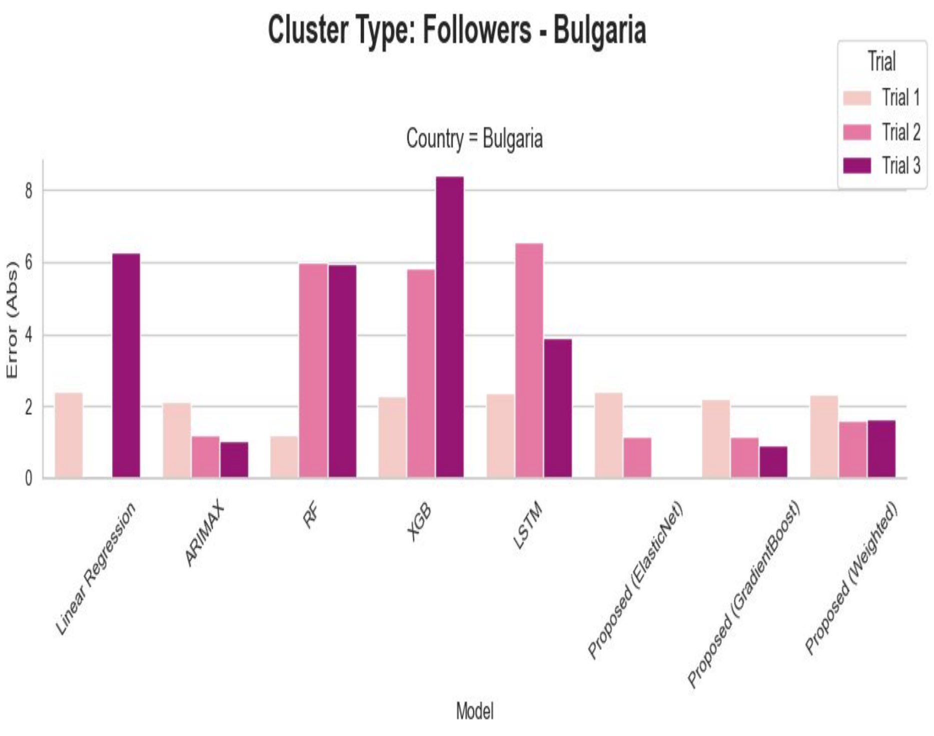


MedAE Visualisation for Leader Cluster



Outliers in the clusters





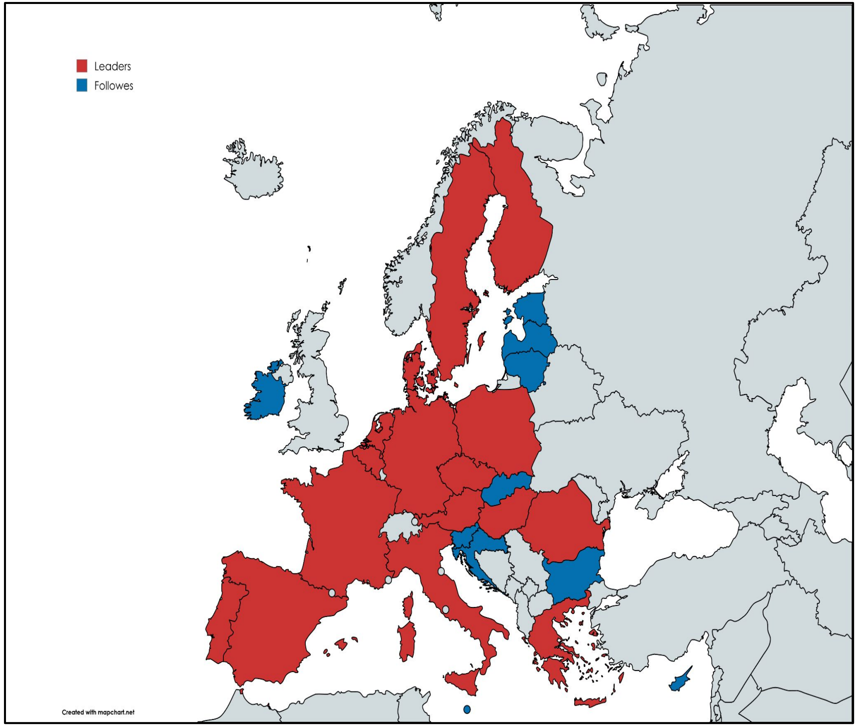
2025 CASE STUDY

- Mirrors Trial 3 configuration but **retrained on full 2020 - 2024 data** (includes 2024 which was test set in Trial 3)

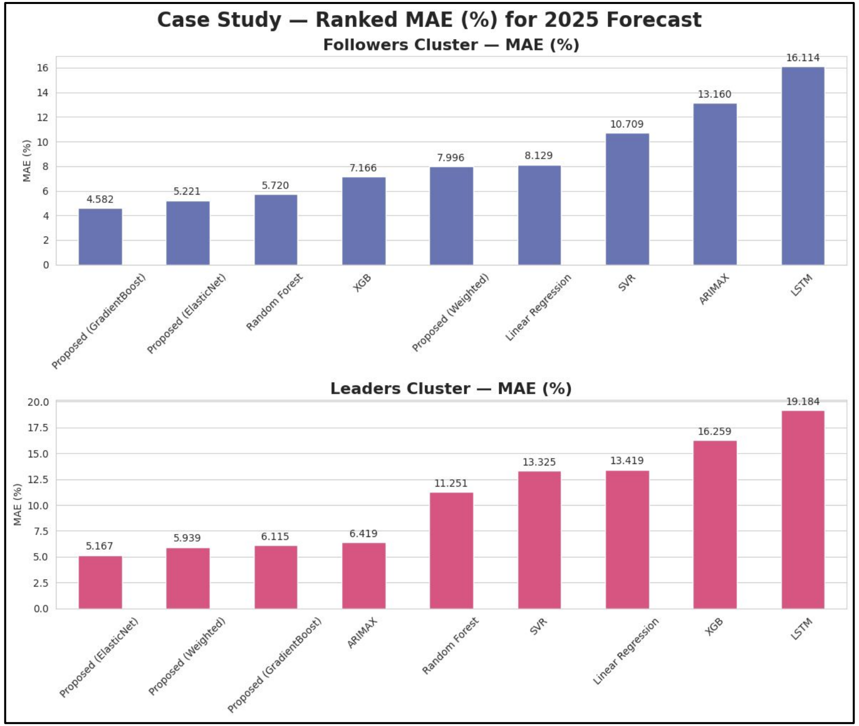
- Purpose: validate whether the proposed framework holds **beyond the original test horizon** under real-world conditions

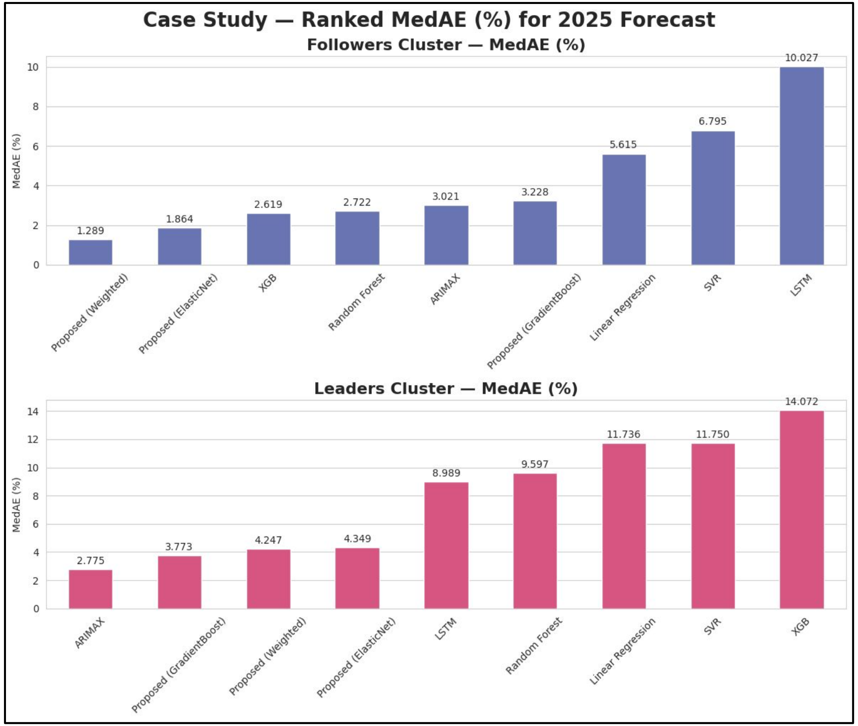
- All 9 models compared to provide comprehensive benchmark

- **Cluster update:** Greece transitions from Followers → **Leaders** in 2025 clustering → **16 Leaders, 11 Followers**



MAE and MedAE values for Case Study 2025 forecast





Future Outlook & Future Research Direction

- **Higher-frequency data**: Monthly/quarterly observations would massively increase model's training sample size.

- **Policy variable integration**: Harmonized, time-varying database of purchase subsidies, tax exemptions, and registration fees across all 27 EU member states.

- **Adaptive Country Segmentation:** Continuously update country cluster memberships annually as new data arrive.

- **Broader applicability**: Apply framework to G20 economies or European Economic Area (including Norway, UK, Iceland) to test cross-regional generalizability; Norway has the world's highest BEV share and represents a uniquely informative additional market.

THANK YOU

Questions?

Mathematical Formulations: Missing Data

- **Log-Linear Interpolation**

$$\hat{y}_t = \exp\left(\log(y_{t_1}) + \frac{t - t_1}{t_2 - t_1}\left(\log(y_{t_2}) - \log(y_{t_1})\right)\right)$$

Where t_1 and t_2 are nearest observed time points (here, years) on either side of the missing value.

- **CAGR-Based Extrapolation:**

CAGR over a period of n years is:

$$CAGR = \left(\frac{y_T}{y_{T-n}}\right)^{\frac{1}{n}} - 1 \xrightarrow[\text{missing year}]{\text{Extrapolated values for the}} \hat{y}_{T+1} = y_T \times (1 + CAGR)$$

here, y_T is the recent year and y_{T-n} is the observed values in the n years prior.

Mathematical Formulations: Transformations

- **Logarithmic Transformations for right-skewed variables**

For variables y>0, logarithm transformation is as follows:

$$y^* = \log(y) \text{ — (1)} \qquad\qquad y^* = \log(1+y) \text{ — (2)}$$

Where, (2) is when variables have zero values. Here, for example, in countries with no recorded registrations or numbers in the initial years.

Then the predictions are back-transformed to the original scale using exponential function.

$$\hat{y} = \exp(\hat{y}^*) \text{ — (3)}$$

Mathematical Formulations: Transformations

- **Yeo-Johnson Transformation for left-skewed variables**

$$\psi(y,\lambda) = \begin{cases} \dfrac{(y+1)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0,\ y \geq 0 \\ \log(y+1) & \text{if } \lambda = 0,\ y \geq 0 \\ \dfrac{-((-y+1)^{2-\lambda} - 1)}{2-\lambda} & \text{if } \lambda \neq 2,\ y < 0 \\ -\log(-y+1) & \text{if } \lambda = 2,\ y < 0 \end{cases}$$

Here, λ is the transformation parameter to make the transformed data as close to normal as possible.

Pearson Correlation

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

where, X’ and Y’ are the sample mean of X and Y, respectively.

The coefficient *r* takes values in the range [−1, 1],

where *r* = 1 indicates a perfect positive linear relationship,

r = −1 indicates a perfect negative linear relationship,

r = 0 indicates the absence of any linear association

VIF

Checks “If this feature at question is giving new information or just repeating information already present? => This creates **multicollinearity.**

The VIF of a predictor X_j is defined as:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where, R^2_j ($0 <= R^2_j <= 1$) iis the coefficient of determination obtained by regressing Xj on all other predictor variables in the dataset.

Mathematical Formulations: PCA

Let X be a n x p matrix where n are the observations and p are the standardised features with a covariance matrix Σ.

The Eigen value decomposition:

$$\Sigma \mathbf{w}_j = \lambda_j \mathbf{w}_j, \quad j = 1, 2, \ldots, p$$

where, w_j are the corresponding eigenvectors and λ_j are the eigenvalues such that $\lambda_1 > \lambda_2 > \lambda_1 > \lambda_3 > \ldots > \lambda_p \geq 0$.

and the j-th principle component (projection of data onto that direction) is

$$\mathbf{z}_j = \mathbf{X}\mathbf{w}_j$$

Mathematical Formulations: PCA

- Explained variance by the j-th component is

$$\text{Explained Variance}_j = \frac{\lambda_j}{\sum_{k=1}^{p} \lambda_k}$$

- Loading factor of a variable X_k on principle component j is given by the formula

$$l_{kj} = w_{kj}\sqrt{\lambda_j}$$

- The critical prerequisite of PCA is standardisation. For a variable X^*_j standardisation is

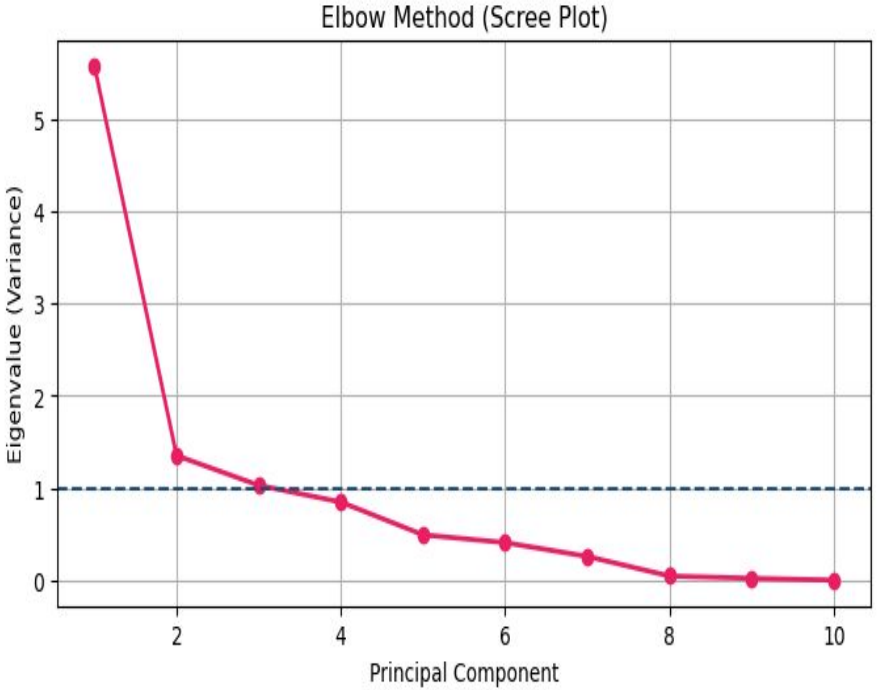
$$X_j^* = \frac{X_j - \bar{X}_j}{s_j}$$

where, $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation of variable j.

Loading Factors & Elbow Method for PCs in Trial 1

Feature	PC1	PC2	PC3
Log Recharging Points	0.229	0.555	0.085
Real Range	0.410	-0.038	-0.001
Log Available BEV Models	0.404	-0.009	0.148
YJ Purchase Price (EUR)	0.343	0.047	0.389
AC Recharging Speed (km/h)	0.362	-0.017	-0.022
Battery Capacity	0.396	-0.081	-0.210
Log DC Recharging Speed (km/h)	0.383	-0.090	-0.286
Employment Growth	0.186	-0.237	0.524
Log GDP	0.038	0.755	-0.160
Log CPI	-0.164	0.219	0.627

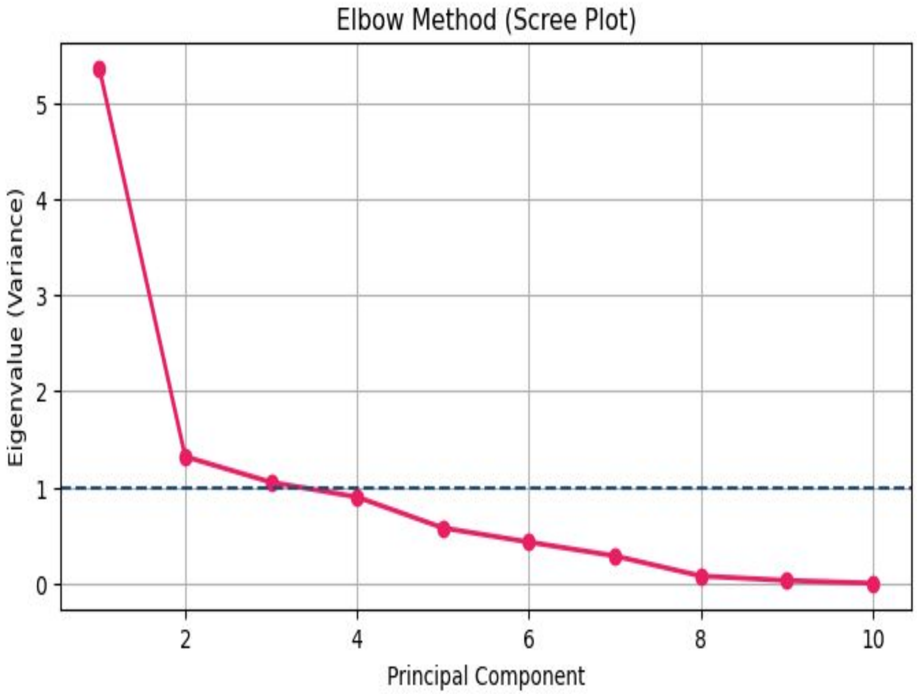
Table 4.4: PCA Loading Factors — Trial 1 (2011–2019 Training Dataset).



Loading Factors & Elbow Method for PCs in Trial 2

Feature	PC1	PC2	PC3
Log Recharging Points	0.294	0.446	-0.031
Real Range	0.420	-0.021	-0.030
Log Available BEV Models	0.396	0.064	0.107
YJ Purchase Price (EUR)	0.331	-0.161	0.058
AC Recharging Speed (km/h)	0.325	-0.193	-0.217
Battery Capacity	0.409	-0.061	-0.138
Log DC Recharging Speed (km/h)	0.404	-0.010	-0.072
Employment Growth	0.164	-0.385	0.391
Log GDP	0.059	0.720	-0.143
Log CPI	0.070	0.254	0.859

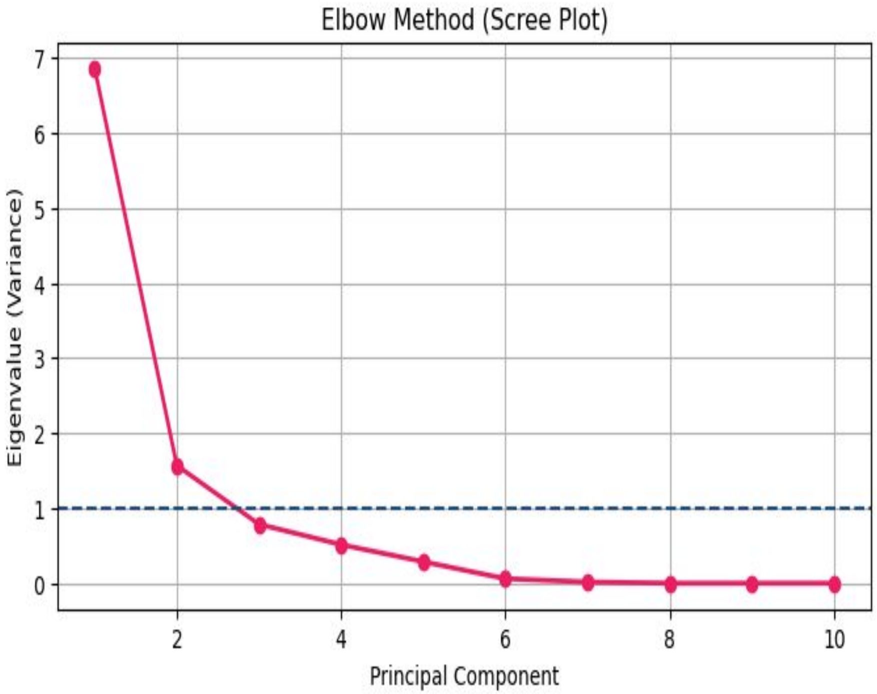
Table 4.7: PCA Loading Factors — Trial 2 (2011–2022 Training Dataset).



Loading Factors & Elbow Method for PCs in Trial 3

Feature	PC1	PC2
Log Recharging Points	-0.132	-0.644
Real Range	-0.381	0.029
Log Available BEV Models	-0.381	0.028
YJ Purchase Price (EUR)	-0.380	0.021
AC Recharging Speed (km/h)	-0.336	-0.009
Battery Capacity	-0.381	0.026
Log DC Recharging Speed (km/h)	-0.381	0.034
Employment Growth	-0.218	0.264
Log GDP	-0.038	-0.715
Log CPI	-0.310	0.016

Table 4.10: PCA Loading Factors — Trial 3 (2020–2023 Training Dataset).



Mathematical Formulations: Selecting #Optimal clusters

- Elbow Method:

$$k^* = \arg\max_k \left(\frac{d^2 \mathrm{WCSS}}{dk^2} \right)$$

Here, **WCSS stands for Within-Cluster Sum of Squares**, *d* is the derivative and *k* = 1,2,3…

In other words, the elbow is the value of *k* that maximises the second derivative of the WCSS curve with respect to *k*.

The target is to find the optimal value of *k*. This happens at the point where the WCSS curve bends sharply forming an elbow like curve beyond which the marginal reduction in WCSS no longer justifies the additional complexity of an extra cluster.

Mathematical Formulations: Selecting #Optimal clusters

- Silhouette Method was a validation step

For observation x_i assigned to cluster C_j, the silhouette coefficient s(i) is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where, a(i) is the mean distance between xi and all other observations in the same cluster (intra-cluster cohesion).

$$a(i) = \frac{1}{|C_j| - 1} \sum_{\mathbf{x}_l \in C_j, l \neq i} \|\mathbf{x}_i - \mathbf{x}_l\|$$

and b(i) is the mean distance between xi and all observations in the nearest neighboring cluster (inter-cluster cohesion).

$$b(i) = \min_{l \neq j} \frac{1}{|C_l|} \sum_{\mathbf{x}_m \in C_l} \|\mathbf{x}_i - \mathbf{x}_m\|$$

Cluster centroids in the principal component space for Trial 1

Period	Cluster	PC1	PC2	PC3
Pre-COVID	Cluster 0 (Leaders)	0.114	0.811	-0.090
Pre-COVID	Cluster 1 (Followers)	-0.166	-1.180	0.131
Post-COVID	Cluster 0 (Followers)	4.882	-0.293	0.741
Post-COVID	Cluster 1 (Leaders)	5.093	1.677	0.470

Table 4.5: Cluster Centroids in PCA Space - Pre-COVID and Post-COVID Periods (Trial 1).

Cluster centroids in the principal component space for Trial 2

Period	Cluster	PC1	PC2	PC3
Train (2011-2022)	Cluster 0 (Leaders)	0.156	0.781	-0.126
Train (2011-2022)	Cluster 1 (Followers)	-0.196	-0.982	0.138
Test (2023-2024)	Cluster 0 (Leaders)	5.502	0.929	-0.269
Test (2023-2024)	Cluster 1 (Followers)	5.192	-0.924	0.115

Table 4.8: Cluster Centroids in PCA Space - Training and Test Periods (Trial 2).

Cluster centroids in the principal component space for Trial 3

Period	Cluster	PC1	PC2
Train (2020–2023)	Cluster 0 (Leaders)	-0.078	-0.935
Train (2020–2023)	Cluster 1 (Followers)	0.097	1.169
Test (2024)	Cluster 0 (Leaders)	-4.151	0.919
Test (2024)	Cluster 1 (Followers)	-4.334	-1.200

Table 4.11: Cluster Centroids in PCA Space – Training and Test Periods (Trial 3).

Mathematical Background: How the dataset looks like?

Country	Year	PC1	PC2	PC3	Log_BEV%
A	2011	0.31	1.20	-0.40	1.85
B	2011	-0.95	-0.95	0.12	0.08
C	2011	0.28	1.15	-0.35	2.10
…	…	…	…	…	…

Input features

Target

Mathematical Background: Linear Regression

For *n* observations and *p* predictor variables, the model is defined as:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

where, $y \in R^n$ is the vector of target observations

$X \in R^{n \times (p+1)}$ is the design matrix containing the predictor variables

$\beta \in R^{p+1}$ is the vector of regression coefficients

$\varepsilon \in R^n$ is the error term.

Then the regression will be written as -

$$\text{Log\_BEV\%} = \beta_0 + \beta_1 \cdot PC1 + \beta_2 \cdot PC2 + \beta_3 \cdot PC3 + \varepsilon$$

where, β_0 = intercept

β_1, β_2, β_3= regression coefficients associated with each principal components

Mathematical Background: Linear Regression

We have n=243 observations for 27 countries and 9 years in T1 (which can change depending on the trial), so the design matrix and target vector looks like -

$$X = \begin{bmatrix} 1 & PC1_1 & PC2_1 & PC3_1 \\ 1 & PC1_2 & PC2_2 & PC3_2 \\ 1 & PC1_3 & PC2_3 & PC3_3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & PC1_{243} & PC2_{243} & PC3_{243} \end{bmatrix} \qquad y = \begin{bmatrix} \text{Log\_BEV\%}_1 \\ \text{Log\_BEV\%}_2 \\ \text{Log\_BEV\%}_3 \\ \vdots \\ \text{Log\_BEV\%}_{243} \end{bmatrix}$$

The Ordinary Least Square (OLS) estimator minimised the sum of squared residuals by choosing the best β:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\beta\|^2 = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y}$$

provided that $X^T X$ is invertible

Mathematical Background: Support Vector Machine

SVR is the entended version of Linear Regression. Here, the function takes the form:

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$

where *w* is the weight vector and *b* is the bias term. The flatness of *f* is controlled by minimising the $||w||^2$ and the *ε*-insensitive loss function is defined as

$$\underset{\mathbf{w},b,\xi,\xi^*}{\text{minimize}} \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)$$

subject to:

$$\begin{cases} y_i - \mathbf{w}^\top \mathbf{x}_i - b \leq \varepsilon + \xi_i \\ \mathbf{w}^\top \mathbf{x}_i + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases}$$

where *C* >0 is the regualrisation parameter which controls the flatness of *f* and the tolerance of training errors exceeding *ε*. ξ_i and ξ_i^* are slack variables for violations above and below the tube respectively.

Mathematical Background: Support Vector Machine

Kernel Trick and RBF Kernel

SVR employs the kernel trick to implicitly map the input features into a higher-dimensional feature space where a linear function can capture non-linear relationships in the original input space.

The kernel function $K(x_i, x_j)$ computes the inner product of two observations in the transformed feature space.

In this study, Radial Basis Function (RBF) was used and it is defined as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right)$$

where $\gamma>0$ controls the width of the width of the kernel and determines the reach of each training observation's influence.

Mathematical Background: Random Forest

Let say for T=100 trees, each tree f_t is trained on a bootstrapped sample D_t drawn with replacement from the training data D:

$$D_t = \{(x_i^*, y_i^*)\}_{i=1}^{n} \; ; \; (x_i^*, y_i^*) \sim D \text{ with replacement}$$

At each node, only m = |sqrt(p)| features randomly selected as split candidates.

For regression tasks, the prediction of the new observation is the average of the predictions of all tress T.

$$\hat{f}_{RF}(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} \hat{f}_t(\mathbf{x})$$

Mathematical Background: XGBoost

XGBoost is a scalable and regularized gradient boosting framework. Gradient Boosting builds up an ensemble of decision trees sequentially, with each tree trained to correct the residual errors of the cumulative ensemble from the previous iteration. The prediction of a gradient boosting model with *T* trees is:

$$\hat{y}_i = \sum_{t=1}^{T} f_t(\mathbf{x}_i), \quad f_t \in \mathcal{F}$$

where *F* is the space of regression trees. At each iteration t, a new tree ft is added to minimize the regularized objective:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t)$$

where l is a differentiable loss function and $\Omega(f_t)$ is the regularization term defined as:

$$\Omega(f_t) = \gamma T_t + \frac{1}{2}\lambda \left\|\mathbf{w}_t\right\|^2$$

T_t is the number of leaves in tree t, w_t is the vector of leaf weights, γ penalizes tree complexity, and λ penalizes large leaf weights.

Mathematical Background: ARIMAX

ARIMAX(p, d, q) model is defined as

$$\phi(B)(1-B)^d y_t = \beta^\top \mathbf{x}_t + \theta(B)\varepsilon_t$$

where, B is the backshift operator = $By_t = y_{t-1}$
$\Phi(B) = 1 - \Phi_1(B) - \dots - \Phi_p(B)^p$ (autoregressive polynomial)
$\theta(B) = 1 + \theta_1(B) + \dots + \theta_q(B)^q$ (moving average polynomial)
β = is the vector of coefficients for the exogenous predictors
x_t = vector of PCA scores (exogenous inputs)
d = 1 (1st derivative): $\triangle y_t = y_t - y_{t-1}$
$\varepsilon_t \sim N(0, \sigma^2)$ is white noise

Mathematical Background: Selection of *p* and *q* for ARIMAX

AIC for order selection

$$AIC = 2k - 2\ln(\hat{L})$$

where, *k* is the number of estimated parameters and *L* is the maximized value of the likelihood function of the model.

Mathematical Background: LSTM

Forget gate: determines what proportion of the previous cell state to retain:

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f\right) \tag{2.36}$$

Input gate: determines what new information to write to the cell state:

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i\right) \tag{2.37}$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\right) \tag{2.38}$$

Cell state update:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{2.39}$$

Output gate: determines what information from the cell state to expose as the hidden state:

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o\right) \tag{2.40}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{2.41}$$

where $\odot$ denotes the Hadamard (element-wise) product, σ is the sigmoid activation function, $\tanh$ is the hyperbolic tangent activation function, and $\mathbf{W}_f, \mathbf{W}_i, \mathbf{W}_c, \mathbf{W}_o$ and $\mathbf{b}_f, \mathbf{b}_i, \mathbf{b}_c, \mathbf{b}_o$ are the learned weight matrices and bias vectors for each gate respectively [HS97].

Mathematical Background: Stacking Ensemble Base Learners

For M base learners $\{f_1, f_2, \ldots, f_M\}$ trained on a training dataset D_{train}, the stacking procedure generates a meta-feature matrix

$$\tilde{\mathbf{X}} = \begin{bmatrix} f_1(\mathbf{x}_1) & f_2(\mathbf{x}_1) & \cdots & f_M(\mathbf{x}_1) \\ f_1(\mathbf{x}_2) & f_2(\mathbf{x}_2) & \cdots & f_M(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(\mathbf{x}_n) & f_2(\mathbf{x}_n) & \cdots & f_M(\mathbf{x}_n) \end{bmatrix}$$

where each column contains the predictions of one base learner (in this study M=2) across all observations. The meta-learner *g* is then trained on *X* with the true target values y as the response:

$$\hat{y} = g(\tilde{\mathbf{x}}) = g(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_M(\mathbf{x}))$$

Mathematical Background: Stacking Ensemble Meta Learners

- Weighted Average

$$\hat{y} = w \cdot \hat{y}_{ARIMAX} + (1 - w) \cdot \hat{y}_{LSTM}$$

where $w \in [0,1]$ is the weight assigned to the ARIMAX prediction and 1-w is the complementary weight assigned to the LSTM prediction.

The optimal weight is determined by minimizing the Mean Squared Error of the combined forecast on the training data:

$$w^* = \underset{w \in [0,1]}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - w\hat{y}_{ARIMAX,i} - (1 - w)\hat{y}_{LSTM,i}\right)^2$$

Mathematical Background: Stacking Ensemble Meta Learners

- Elastic Net Regression is a regularized regression method that combines the L1 penalty of the Lasso with the L2 penalty of Ridge regression:

$$\hat{\beta}_{EN} = \underset{\beta}{\operatorname{argmin}} \left\| \mathbf{y} - \tilde{\mathbf{X}}\beta \right\|^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$$

where λ_1 and λ_2 control the strength of the L1 and L2 penalties respectively.

L1 penalty encourages sparsity which drives some coefficients to exactly zero.

L2 penalty stabilizes the solution when predictors are correlated.

Mathematical Background: Stacking Ensemble Meta Learners

- Gradient Boost similar to XGBoost, builds an additive ensemble of weak learners that typically gives shallow decision trees by sequentially fitting each new learner to the residual errors of the cumulative ensemble from the previous iteration.

$$\hat{y} = \sum_{t=1}^{T} \nu \cdot h_t(\tilde{\mathbf{x}})$$

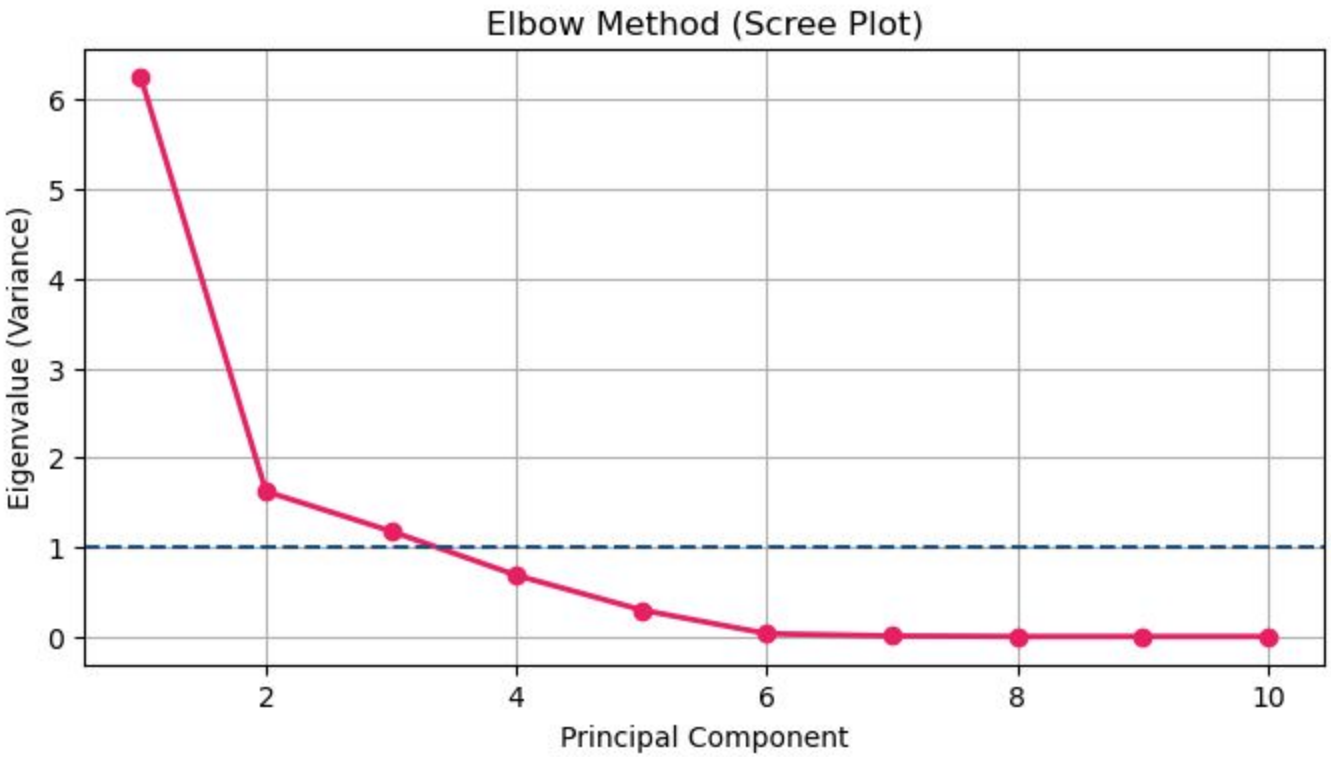
where h_t is the t-th weak learner fitted to the residuals of the previous ensemble, $v \in (0, 1]$ is the learning rate that controls the contribution of each tree, and $x = [y^{ARIMAX}, y^{LSTM}]$ is the two-dimensional meta-feature vector of base learner predictions.

Why not other models as base learners?

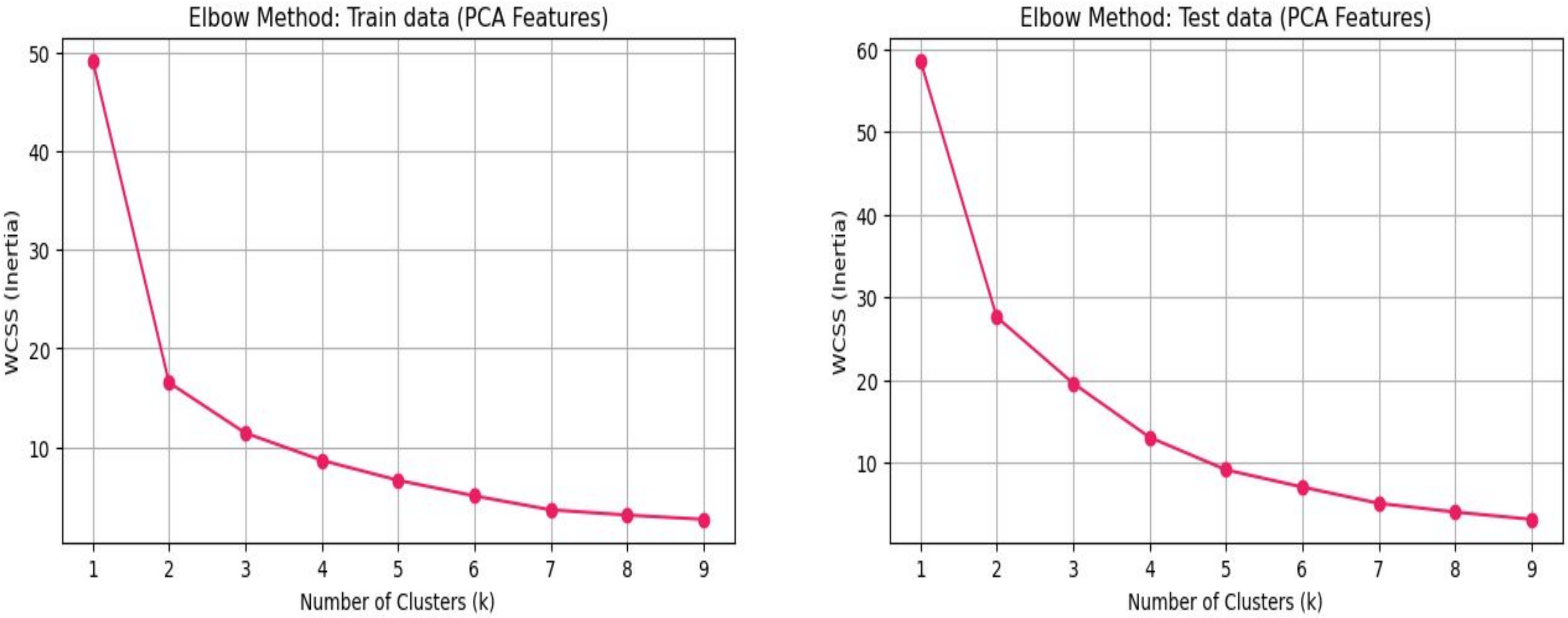
1. **Random Forest & XGBoost:**
   a. Makes similar types of errors, because both of them failed when the post-COVID values exceed the Pre-COVID values.
   b. Cannot extrapolate outside the training data.
   c. Their error patterns correlate => combining them cancels out nothing.

2. **SVR**
   a. SVR failed catastrophically even as a standalone model (MAE = 150.71% in Trial 1 Leaders). A base learner that performs this badly would add noise to the ensemble rather than useful signal. The meta-learner would simply learn to give SVR zero weight making it pointless to include.

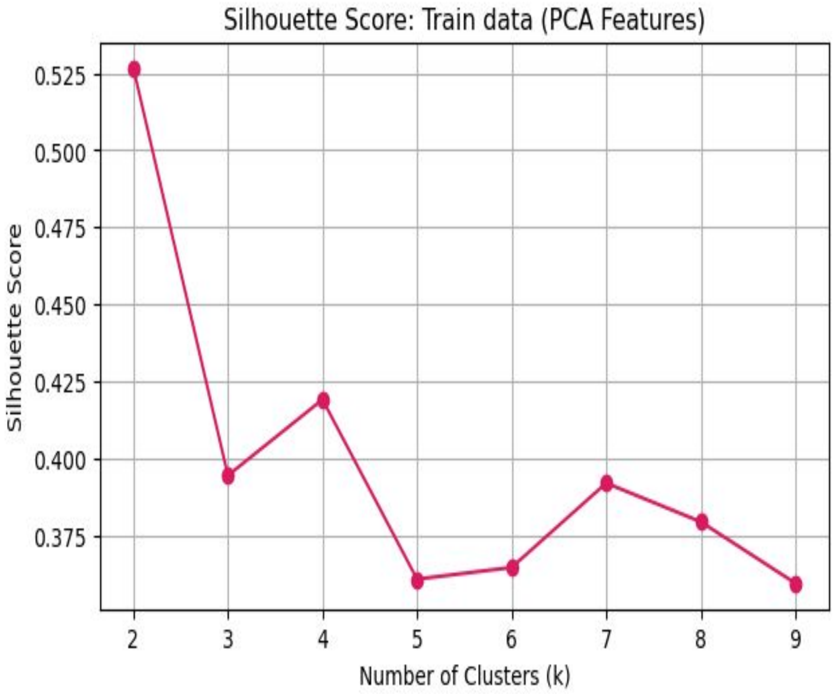
Case Study: Scree plot to determine the #principal components

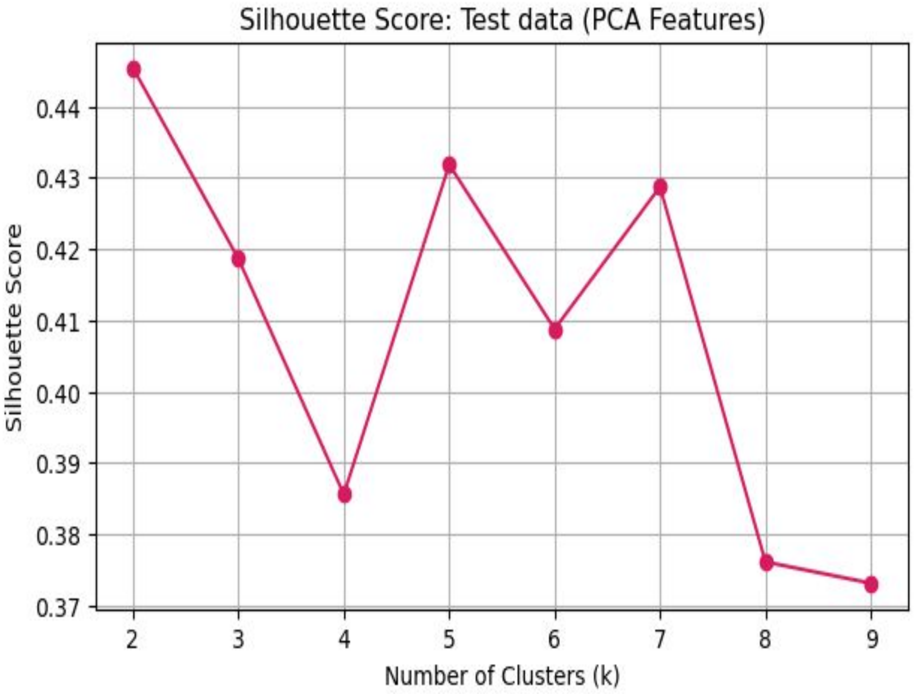


Case Study: Elbow method to determine the #clusters



Case Study: Silhouette method to determine the #clusters





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